ZOO501
PAST PAPERS
FINAL TERMS

COLLECTED BY
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Exon Intron?

Both introns and exons are the genetic material. But the essential difference between the two is that one is made into protein and the other is not. DNA molecules are made up of small molecules that are called nucleic acids. An exon is a sequence of nucleic acids that are represented in the mRNA molecule.

How to avoid polyspermy?

prevent the entry of more than one sperm into the egg. The sea urchin egg has two mechanisms to avoid polyspermy: a fast reaction, accomplished by an electric change in the egg plasma membrane, and a ***slower reaction,*** caused by the exocytosis of the cortical granules

Defect of neurulation?

Neural tube defects (NTDs) are severe congenital malformations affecting 1 in every 1000 pregnancies. 'Open' NTDs result from failure of primary neurulation as seen in anencephaly, myelomeningocele (open spina bifida) and craniorachischisis.

Meroblastic cleavage?

meroblastic cleavage, wherein the early cell divisions do not cut through the yolk of the egg. These cells form a blastoderm.

Fraternal and Identical twin?

Twins are two offspring produced by the same pregnancy. Twins can be either monozygotic(**'identical'**), meaning that they develop from one zygote, which splits and forms two embryos, or dizygotic (**'fraternal'**), meaning that each twin develops from a separate egg and each egg is fertilized by its own sperm cell.

What is blastocyst?

Blastocyst: A thin-walled hollow structure in early embryonic development that contains a cluster of cells called the inner cell mass from which the embryo arises.

How acrosomal reaction initiate in sea urchin?

In the sea urchin, contact with egg jelly initiates the acrosome reaction, which is a calcium mediated process.

The acrosomal vesicle (green) fuses with the plasma membrane, releasing enzymes from the tip of the sperm that aid digestion of egg jelly.

How genetic material exchange in paramecium?

The micronucleus is the generative, or germline nucleus, containing the genetic material that is passed along from one generation to the next. Paramecium reproduces asexually, by binary fission. During reproduction, the macronucleus splits by a type of amitosis, and the micronuclei undergo mitosis.

What is fate of metencephalon?

The fate of the brain vesicles is determined by the combination of transcription factors. Signals from the secondary organizer, the anterior neural ridge and the mesencephalon–metencephalon boundary (isthmus), regulate expression of the transcription factors in the adjacent region and organize its fate.

.Function of Placenta?

The placenta acts to provide oxygen and nutrients to the fetus, whilst removing carbon dioxide and other waste products. It metabolises a number of substances and can release metabolic products into maternal and/or fetal circulations.

. What is campomelic dysplasia?

Campomelic dysplasia is a severe disorder that affects development of the skeleton, reproductive system, and other parts of the body. This condition is often life-threatening in the newborn period. The term "**campomelic**" comes from the Greek words for "bent limb."

.Two different pathways of neural crest cell?

Neural crest cells that become the pigmentsynthesizing melanocytes migrate dorsolaterally into the ectoderm and continue on their way toward the ventral midline of the belly.

The second migratory pathway takes the trunk neural crest cells ventrolaterally through the anterior half of each Sclerotic

.Myogenesis

Myogenesis: The Development of Muscle Paracrine factors instruct the Myotome cells to become muscles by inducing them to synthesize the MyoD protein. Factors from the surrounding environment induce the Pax3 transcription factor. In the absence of other inhibitory transcription factors (such as those the found in the sclerotome cells), Pax3 then activates the genes encoding two muscle-specific transcription factors, Myf5 and MyoD. MyoD and Myf5 belong to a family of transcription factors called the myogenic bHLH (basic helix-loop-helix) proteins

(sometimes also referred to as the MyoD family). The proteins of this family all bind to similar sites on the DNA and activate muscle-specific genes. MyoD also directly activates its own gene. While Pax3 is found in several other cell types, the myogenic bHLH proteins are specific for muscle cells. Any cell making a myogenic bHLH transcription factor such as MyoD or Myf5 is committed to becoming a muscle cell. Transfection of genes encoding any of these myogenic proteins into a wide range of cultured cells converts those cells into muscle

Myogenesis is the formation of muscular tissue, particularly during embryonic development. Muscle fibers form from the fusion of myoblasts into multi-nucleated fibers called myotubes. ... In this stage, calcium ions are critical for development.

.Primary and Secondary neurulation?

In primary neurulation, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.

In secondary neurulation, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube. Neurulation in fishes is exclusively secondary

Cortical granule reaction?

Cortical granules are membrane-bound structures containing proteolytic enzymes.
However, whereas each sperm contains one acrosomal vesicle, each sea urchin egg contains approximately 15,000 cortical granules

.Embryonic Shield?

Also called embryonic shield. in the early embryo of mammals, the flattened inner cell mass that arises at the end of the blastocyst stage and from which the embryo begins to differentiate.

.Slow block polyspermy?

The slow block to polyspermy begins within 10 seconds of fusion of the sperm and egg plasma membranes.

A compound called inositol triphosphate (IP3) causes the release of Ca++ from intracellular stores in the egg endoplasmic reticulum.

.Osteogenesis?

Osteogenesis imperfecta (OI), also known as brittle bone disease, is a group of genetic disorders that mainly affect the bones. It results in bones that break easily.

.Types of Cleavage?

the entire cell is divided equally. Four major holoblastic cleavage types can be observed in general: radial,

spiral, bilateral, and rotational. Egg cells that have larger quantities of yolk undergo meroblastic cleavage after fertilization, in which only a portion of the zygote undergoes cleavage.

.Response from egg to sperm?

The responses of the egg to the sperm can be divided into "early" responses, which occur within seconds of the cortical reaction, and "late" responses, which take place several minutes after fertilization begins.

.Blastocoel function?

The blastocoel probably serves two major functions in frog embryos:

(1) it permits cell migration during gastrulation, and

(2) it prevents the cells beneath it from interacting prematurely with the cells above it.

.Tunicate gastrulation?

Tunicates, like sea urchins and vertebrates, follow the deuterostome pattern of gastrulation in which the blastopore becomes the anus.Tunicate gastrulation is characterized by the invagination of the endoderm, the involution of the mesoderm, and the epiboly of the ectoderm

.Stereoblastula?

Blastulae produced by radial cleavage have no blastocoel and are called stereoblastulae.

Neural Tube Close?

The failure of the entire neural tube to close over the entire body axis is called Craniorachischisis. Collectively, neural tube defects are not rare in humans, as they are seen in about 1 in every 500 live births.

.Function of Enhancer?

Enhancer function underlies regulatory processes by which cells establish patterns of gene expression. Recent results suggest that enhancers are specified by particular chromatin marks in pluripotent cells, which may be modified later in development to alter patterns of gene expression and cell differentiation choices.

.Composition of hyaline membrane in Sea Urchin?

The hyaline layer is actually made up of two layers, an outer lamina made primarily of hyalin protein and an inner lamina composed of fibropellin proteins

.Blastocoel Function?

The blastocoel probably serves two major functions in frog embryos

: (1) it permits cell migration during gastrulation, and

(2) it prevents the cells beneath it from interacting prematurely with the the cells above it.

.Blastomere cleavage?

Production of a multicellular organism, many cells must be produced from the singlecelled zygote. This task is accomplished by cleavage, a series of consecutive cell divisions. Cells produced during cleavage are called blastomeres.

Primary and Secondary Neural tube?

The neural plate folds in upon itself to form the neural tube, which will later differentiate into the spinal cord and the brain, eventually forming the central nervous system. Different portions of the neural tube form by two different processes, called primary and secondary neurulation, in different species.

.What is Blastocyst?

A mammalian blastula in which some differentiation of cells has occurred.

How acrosomal reaction initiate in Sea Urchin?

The acrosomal reaction in sea urchins is initiated by contact of the sperm with the egg jelly. Contact with egg jelly causes the exocytosis of the sperm's acrosomal vesicle and the release of proteolytic enzymes that can digest a path through the jelly coat to the egg surface.

.What is Fete metacephone?

.What is meroblastic Cleavage?

Depending mostly on the amount of yolk in the egg, cleavage can be holoblastic (complete or total) or meroblastic (partial). In meroblastic cleavage, only a portion of the egg divides. This is because of the large

amount of yolk in the egg cell that the cleavagefurrows do not penetrate the yolk. ... super ficiall cleavage.

Write 2 major difference pathway of nueral crest cells?

The trunk neural crest, whose cells take one of two major pathways.

Neural crest cells that become the pigmentsynthesizing melanocytes migrate dorsolaterally into the ectoderm and continue on their way toward the ventral midline of the belly. The second migratory pathway takes the trunk neural crest cells ventrolaterally through the anterior half of each sclerotome. Sclerotomes are blocks of mesodermal cells, derived from somites, that will differentiate into the vertebral cartilage of the spine.

The vagal and sacral neural crest, whose cells generate the parasympathetic (enteric) ganglia of the gut. The vagal (neck) neural crest lies opposite chick somites 1-7, while the sacral neural crest lies posterior to somite 28.

The cardiac neural crest is located between the cranial and trunk neural crests. The cardiac neural crest cells can develop into melanocytes, neurons, cartilage, and connective tissue (of the third, fourth, and sixth pharyngeal arches).

.Cortical granule reaction? Cortical reaction?

The cortical reaction is a process initiated during fertilization by the release of cortical granules from the egg, which prevents polyspermy, the fusion of multiple sperm with one egg.

Cleavage pattern in sea urchin, nematode and tunicate?

In all four invertebrates described here, cleavage is holoblastic. In the sea urchin,cleavage is radial; in the snail, spiral; in the tunicate, bilateral; and in the nematode, rotational. ... In some, such as the sea urchin and tunicate, the axes are established at fertilization through determinants in the egg cytoplasm.

.How secondary hypoblast form in avians?

Gastrulation of the Avian Embryo Shortly thereafter, a sheet of cells from the posterior margin of the blastoderm (distinguished from the other regions of the margin by Koller's sickle, a local thickening) migrates anteriorly to join the polyinvagination islands, thereby forming the secondary hypoblast.

.Difference in cotice and nuclei?

.Larval Stage of Frog?

A tadpole (also called a pollywog) is the larval stage in the life cycle of an amphibian, particularly that of a frog or toad. They are usually wholly aquatic, though some species have tadpoles that are terrestrial.

What happened neural tube fail to fuse?

Neural tube defects—Spina bifida, Anencephaly and Encephalocele. ... The neural tube goes on to develop into your baby's brain and spinal cord. When the neural tube does not fuse together properly a problem with the development of the spine or brain occurs. This is known as a neural tube defect.

.Resting potential membrane?

The opening of the calcium channels involve the receptor's activating a cation channel (for sodium, potassium, or calcium), which would change the resting potential of the sperm plasma membrane. The calcium channels in the membrane would be sensitive to this change in membrane potential, allowing calcium to enter the sperm.

.Neural plate?

The neural plate is a key developmental structure that serves as the basis for the nervous system. Opposite the primitive streak in the embryo, ectodermal tissue thickens and flattens to become the neural plate. The region anterior to the primitive knot can be generally referred to as the neural plate.

.Event of Fertilization?

The result of fertilization is a cell (zygote) capable of undergoing cell division to form a new individual. The fusion of two gametes initiates several reactions in the egg. One of these causes a change in the egg membrane(s), so that the attachment of and penetration by more than one spermatozoon cannot occur.

. How Discovered sperm?

A.V Leeuwenhoek, the **Dutch** microscopist who co-discovered sperm in 1678, first believed them to be parasitic animals living within the semen (hence the term spermatozoa, meaning "sperm animals").

Conditional SP with example?

Conditional specification: o Characteristic of all vertebrates and few invertebrates. o Specification by interactions between cells. Relative positions are important

Tunicates gastrulation?

Tunicate gastrulation is characterized by the invagination of the endoderm, the involution of the mesoderm, and the epiboly of the ectoderm. About 4 5 hours after fertilization, the vegetal (endoderm) cells assume a wedge shape, expanding their apical margins and contracting near their vegetal margins.

Ectoderm difference?

the ectoderm differentiates to form the nervous system (spine, peripheral nerves and brain), tooth enamel and the epidermis (the outer part of integument). It also forms the lining of mouth, anus, nostrils, sweat glands, hair and nails

MPF Subunit?

Mitosis-promoting factor contains ***two subunits***. ***The large subunit is called cyclin B. Small subunit of MPF, the cyclin-dependent kinase***

Defect of Neutralization?

Different neural tube defects are caused when various parts of the neural tube fail to close. Failure to close the human posterior neural tube regions at day 27 (or the subsequent rupture of the posterior neuropore shortly thereafter) results in a condition called spina bifida, the severity of which depends on how much of the spinal cord remains exposed. Failure to close the anterior neural tube regions results in a lethal condition, anencephaly. Here, the forebrain remains in contact with the amniotic fluid and subsequently degenerates. Fetal forebrain development ceases, and the vault of the skull fails to form.

Identical Twin and paternalTwin?

To form identical or monozygotic twins, one fertilised egg (ovum) splits and develops into two babies with exactly the same genetic information. To form praternal or dizygotic twins, two eggs (ova) are fertilised by two sperm and produce two genetically unique children.

Difference mono and Polyspermy?

is that monospermy is fertilization of an oocyte by a single spermatozoon while

polyspermy is the penetration of an ovum by more than one sperm.

What is Syncytiotrophoblast?

The syncytiotrophoblast is the primary structure that determines which substances cross the placenta (e.g., nutrients and oxygen) and which substances do not (e.g., maternal hormones and certain toxins).

Anatomical Approach?

The third anatomical approach to developmental biology is teratology, the study of birth defects. ... The fourth anatomical approach is mathematical modeling, which seeks to describe developmental phenomena in terms of equations.

. Mantal and Ventricular zone?

new layer is called the mantle (or intermediate) zone, and the germinal epithelium is now called the ventricular zone (and, later, the ependyma).

Species Specific recognition in sea urchin?

"Species specific recognition of sea urchin gametes occurs at the levels of sperm attraction, sperm activation, and sperm adhesion to the egg surface." The sea-urchinegg jelly contains a species specific chemo-attractant (redact).

How avoid polyspermy?

Slow block of polyspermy

Nevertheless, polyspermy preventing mechanisms are essential in mammals; a secretion reaction, the "cortical reaction" modifies the extracellular coat of the egg (the zona pellucida), and additional mechanisms that are not well understood modify the egg's plasma membrane

Chemotoxix?

Is the movement of an organism in response to a chemical

Type of Egg in insect?

These are megalecithal polylecithal & centrolecithal eggs. The cytoplasm of insect egg is limited to periphery only whole central place is filled by yolk.

Two types of egg envelops are present on egg of insects. Inner - Vitelline membrane - Primary egg membrane.

Two way from neural tube?

There are two major ways of forming a neural tube Primary neurulation Secondary neurulation.

Name of cell movement during gastrulation in vertebrates?

During internalization, endoderm and mesoderm cells move via the blastopore beneath the ectoderm. ...

Patterns of gastrulation cell movements relative to the blastopore and the organizer are similar from fish to mammals, and conserved molecular pathways mediate gastrulation movements.

Vegetal pole and Animal pole?

The yolk-rich pole is referred to as the vegetal pole; the yolk concentration in the animal pole is relatively low.

Campomelic dysplasia?

Campomelic dysplasia (CMD) is a rare genetic disorder characterized by bowing of the long bones and many other skeletal and extraskeletal features. It is frequently lethal in the neonatal period due to respiratory insufficiency, but the severity of the disease is variable, and some patients survive into adulthood.

How egg protected from sunlight?

Many eggs contain ultraviolet filters and DNA repair enzymes that protect them from sunlight

Amniotic Fluid?

The amniotic fluid is the protective liquid contained by the amniotic sac of a gravid amniote. This fluid serves as a cushion for the growing fetus, but also serves to facilitate the exchange of nutrients, water, and biochemical products between mother and fetus.

What create dramatic change in the structure of organism?

Allometric growth can create dramatic changes in the structure of organisms.

Why tunicate give name?

The name, "tunicate" comes from the firm, but flexible body covering, called a tunic. Most tunicates live with the posterior, or lower end of the barrel attached firmly to a fixed object, and have two openings, or siphons, projecting from the other. Tunicates are plankton feeders.

Transcription factor group together?

General transcription factors are involved in the formation of a preinitiation complex. The most common are abbreviated as TFIIA, TFIIB, TFIID, TFIIE, TFIIF, and TFIIH. They

are ubiquitous and interact with the core promoter region surrounding the transcription start site(s) of all class II genes

Difference between Enhancer and Promoter?

An enhancer is a sequence of DNA that functions to enhance transcription. A promoter is a sequence of DNA that initiates the process of transcription. A promoter has to be close to the gene that is being transcribed while an enhancer does not need to be close to the gene of interest.

Polyspermy avoided in sea urchin?

The most common way is to prevent the entry of more than one sperm into the egg. The sea urchin egg has two mechanisms to avoid polyspermy: a fast reaction, accomplished by an electric change in the egg plasma membrane, and a slower reaction, caused by the exocytosis of the cortical granules

Zona pellucida?

The zona pellucida supports communication between oocytes and follicle cells during oogenesis; protects oocytes, eggs, and embryos during development, and regulates interactions between ovulated eggs and free- swimming sperm during and following fertilization.

When Capacitation occur?

The set of physiological changes that allow the sperm to be competent to fertilize the egg is called capacitation.

Capacitation is the penultimate step in the maturation of mammalian spermatozoa and is required to render them competent to fertilize an oocyte. ... In vivo, capacitation occurs after ejaculation, when the spermatozoa leave the vagina and enter the superior female reproductive tract.

What are protostome?

(Greek, "mouth first"), which include the mollusc, arthropod, and worm phyla, are so called because the mouth is formed first, at or near the opening to the gut, which is produced during gastrulation. The anus forms later at another location.

Composition of hyaline membrane sea urchin?

The sea urchin embryo hyaline layer is an extracellular investment which develops within 20 min post insemination of Strongylocentrotus purpuratus eggs and contains a single calciumprecipitable subunit termed hyalin.

How neural groove formed in primary neurulation?

The process begins when the notochord induces the formation of the central nervous system (CNS) by signaling the ectoderm germ layer above it to form the thick and flat neural plate. ... In primary neurulation, the neural plate creases inward until the edges come in contact and fuse.

Syncytium?

A cytoplasm that contains many nuclei is called a syncytium

Demonstration of Sperm enter in the egg Two nuclie?

In 1876 Oscar Hertwig and Herman Fol independently demonstrated sperm entry into the egg and the union of the two cells' nuclei.

. Polyspermy 2 marks

The entrance of multiple sperm polyspermy leads to disastrous consequences in most organisms. In the sea urchin, fertilization by two sperm results in a triploid nucleus, in which each chromosome is represented three times rather than twice (Worse condition). Species have evolved ways to prevent the union of more than two haploid nuclei. The most common way is to prevent the entry of more than one sperm into the egg.

polyspermy describes an egg that has been fertilized by more than one sperm. Diploid organisms normally contain two copies of each chromosome, one from each parent. The cell resulting from polyspermy, on the other hand, contains three or more copies of each chromosome—one from the egg and one each from multiple sperm. Usually, the result is an inviable zygote

. Zona pellucida

Zona pellucida: The strong membrane that forms around an ovum as it develops in the ovary. The membrane remains in place during the egg's travel through the fallopian tube. To fertilize the egg, a sperm must penetrate the thinning zona pellucida. If fertilization takes place, the zona pellucida disappears, to permit implantation in the uterus. The zona pellucida supports communication between oocytes and follicle cells during oogenesis; protects oocytes, eggs, and embryos during development, and regulates interactions between ovulated eggs and free- swimming sperm during and following fertilization.

. Marble bone disease

Osteopetrosis or Marble bone disease

Osteopetrosis is a bone disease that makes bones abnormally dense and prone breakage (fracture). This disorder is characterized by increased bone density and abnormal bone growth.

Marble bone disease, also called osteopetrosis or Albers-Schönberg disease, rare disorder in which the bones become extremely dense, hard, and brittle. The disease progresses as long as bone growth continues; the marrow cavities become filled with compact bone. Because increased bone mass crowds the bone marrow, resulting in a reduced amount of marrow and therefore a reduced capacity to produce red blood cells, severe anemia results

Bone precursor cell

During intramembranous ossification in the skull, neural crest derived mesenchymal cells proliferate and condense into compact nodules. Some of these cells develop into capillaries; others change their shape to become osteoblasts, committed bone precursor cells.

What happens when fusion occur between sperm and egg

Recognition of sperm by the vitelline envelope or zona pellucida is followed by the lysis of that portion of the envelope or zona in the region of the sperm head by the acrosomal enzymes. This lysis is followed by the fusion of the sperm plasma membrane with the plasma membrane of the egg. The sperm and egg plasma membranes then join together, and material from the sperm membrane can later be found on the egg membrane. The sperm nucleus and tail pass through the resulting cytoplasmic bridge, which is widened by the actin polymerization.

The process of fertilization involves a sperm fusing with an ovum. After the sperm enters the cytoplasm of the oocyte (also called ovocyte), the tail and the outer coating of the sperm disintegrate and the cortical reaction takes place, preventing other sperm from fertilizing the same egg. The oocyte now undergoes its second meiotic division producing the haploid ovum and releasing a polar body. The sperm nucleus then fuses with the ovum, enabling fusion of their genetic material. The successful fusion of gametes forms a new organism

Fertilin function

- In mammals, the fertilin proteins in the sperm plasma membrane are essential for sperm membrane- egg membrane fusion
- Mouse fertilin is localized to the posterior plasma membrane of the sperm head. It adheres the sperm to the egg by binding to the 6β1 integrin protein on the egg plasma membrane
- fertilin appears to bind the sperm plasma membrane to the egg plasma membrane and then to fuse the two of them together. When the membranes are fused, the sperm nucleus, mitochondria, centriole, and flagellum can enter the egg
- Fertilin is a sperm surface protein with an essential role in fertilization. It is required for the migration of spermatozoa through the oviduct, for binding to the zona pellucida, and for efficient binding to the egg plasma membrane

Function of dynein

The core of the axoneme consists of two central microtubules surrounded by a row of nine doublet microtubules. The force for sperm propulsion is provided by dynein, a protein that is attached to the microtubules. Dynein hydrolyzes molecules of ATP and can convert the released chemical energy into the mechanical energy that propels the sperm.

Blastomere progress

Blastomeres generally progress through a cell cycle consisting of just two steps:

M (mitosis) and S (DNA synthesis).

Embryonic shield

The cells of both the epiblast and hypoblast intercalate on the future dorsal side of the embryo to form a localized thickening, the embryonic shield.

Also called embryonic shield. in the early embryo of mammals, the flattened inner cell mass that arises at the end of the blastocyst stage and from which the embryo begins to differentiate.

Example of enhancer in human

The human β-globin gene has an enhancer in its 3´ UTR, roughly 700 base pairs

What is homology of Hox genes in droshophila and of mammals

The homology of gene structure and the similarity of expression patterns between Drosophila and mammalian Hox genes suggests that this patterning mechanism is extremely ancient.

Process of neural crest

The neural crest cells. The neural crest cells form in the region that connects the neural tube and epidermis, but then migrate elsewhere; they will generate the peripheral neurons and glia, the pigment cells of the skin, and several other cell types.

The Neural Crest

A group of ectodermal cells that gives rise to the spinal ganglia and various structures of the autonomic nervous system.The neural crest cells originate at the dorsal most region of the neural tube.The neural crest can be divided into four main functional (but overlapping) domains.

The cranial (cephalic) neural crest, whose cells migrate dorsolaterally to produce the craniofacial mesenchyme that differentiates into the cartilage, bone, cranial neurons, glia,and connective tissues of the face.

Major pathways in differentiation of neural crest

The trunk neural crest, whose cells take one of two major pathways.

- Neural crest cells that become the pigment-synthesizing melanocytes migrate dorsolaterally into the ectoderm and continue on their way toward the ventral midline of the belly.

- The second migratory pathway takes the trunk neural crest cells ventrolaterally through the anterior half of each sclerotome.
- Sclerotomes are blocks of mesodermal cells, derived from somites, that will differentiate into the vertebral cartilage of the spine.
- The vagal and sacral neural crest, whose cells generate the parasympathetic (enteric) ganglia of the gut.
- The vagal (neck) neural crest lies opposite chick somites 1-7, while the sacral neural crest lies posterior to somite 28.
- The cardiac neural crest is located between the cranial and trunk neural crests.
- The cardiac neural crest cells can develop into melanocytes, neurons, cartilage, and connective tissue

(of the third, fourth, and sixth pharyngeal arches).

Causes in the activity cyclin of MPF

Mitosis-promoting factor contains two subunits.

The large subunit is called cyclin B.

Small subunit of MPF, the cyclin-dependent kinase

Cyclin B is often encoded by mRNAs stored in the oocyte cytoplasm, and if the translation of this message is specifically inhibited, the cell will not enter mitosis. The presence of cyclin B depends upon its synthesis and its degradation. Cyclin B regulates the small subunit of MPF, the cyclin-dependent kinase.

Difference between Enhancer and Promoter?

An enhancer is a sequence of DNA that functions to enhance transcription. A promoter is a sequence of DNA

that initiates the process of transcription. A promoter has to be close to the gene that is being transcribed while an enhancer does not need to be close to the gene of interest.

Mode of commitment

Three basic modes of commitment are,

- ***Autonomous specification***
- ***Conditional specification***
- ***Syncytial specification***

. What is sea urchin larva name

Pluteus larve

. Cleavage in sea urchin

As the 16-cell embryo cleaves, the eight mesomeres divide to produce two "animal" tiers, an1 and an2, one staggered above the other. The macromeres divide meridionally, forming a tier of eight cells below an2.

How neural groove form

Shortly after the neural plate has formed, its edges thicken and move upward to form the neural folds, while a U-shaped neural groove appears in the center of the plate, dividing the future right and left sides of the embryo.

Retinoic acid foundation

The specification of the ventral neural tube appears to be mediated by external tissues.One agent of ventral specification is the Sonic hedgehog protein, probably originating from the notochord. Another agent specifying the ventral neural cell types is retinoic acid, which probably comes from the adjacent somites.

What is hierarchical development

The developmental hierarchy is the hierarchical series of decisions involving specific preexisting cell types.

These developmental decisions are usually irreversible, that progressively and irreversibly restrict cell fate.

Animal development is the division of embryo into the three germ layers.

1. Ectoderm

2. Mesoderm

3. Endoderm

periosteum

the entire region of calcified spicules becomes surrounded by compact mesenchymal cells that form the **Periosteum (a membrane that surrounds the bone).**

The periosteum is a membrane that covers the outer surface of all bones, except at the joints of long bones. Endosteum lines the inner surface of the medullary cavity of all long bones.

Difference between germ cells and somatic cells

In many species a specialized portion of egg cytoplasm gives rise to cells that are the precursors of the gametes (the sperm and egg). The gametes and their precursor cells are collectively called germ cells, and they are set aside for reproductive function.

All the other cells of the body are called Somatic cells. This separation of somatic cells (which give rise to the individual body) and germ cells (which contribute to the formation of a new generation) is often one of the first differentiations to occur during animal development.

Difference between fraternal or identical twins

Fraternal twins are the result of two separate fertilization events, whereas identical twins are formed from a single embryo whose cells somehow dissociated from one another.

Identical twins occur in roughly 0.25% of human births. About 33% of identical twins have two complete and separate chorions, indicating that separation occurred before the formation of the trophoblast tissue at day 5.

Primary neurulation and secondary neurulation

In primary neurulation, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.

During primary neurulation, the original ectoderm is divided into three sets of cells.

- ***The internally positioned neural tube, which will form the brain and spinal cord.***
- ***The externally positioned epidermis of the skin.***

The neural crest cells. The neural crest cells form in the region that connects the neural tube and

epidermis, but then migrate elsewhere; they will generate the peripheral neurons and glia, the pigment cells of the skin, and several other cell types.

In secondary neurulation, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube. Neurulation in fishes is exclusively secondary.

Secondary neurulation involves the making of a medullary cord and its subsequent hollowing into a neural

tube. In frogs and chicks, secondary neurulation is usually seen in the neural tube of the lumbar (abdominal) and tail vertebrae. In both cases, it can be seen as a continuation of gastrulation. The growing region at the tip

of the lip is called the chordoneural hinge, and it contains precursors for both the posterior most portion of the neural plate and the posterior portion of the notochord. The tip of the tail is the direct descendant of the dorsal blastopore lip, and the cells lining the blastopore form the neurenteric canal.

. Three axes of body foundation

Three axes are the foundations of the body:

- the anterior-posterior axis (head to tail or mouth to anus),
- the dorsal-ventral axis (back to belly),
- the right-left axis (between the two lateral sides of the body).

25. Step of sperm and egg interaction 5 marks

The interaction of sperm and egg generally proceeds according to five basic step:

➢ The chemo-attraction of the sperm to the egg by soluble molecules secreted by the egg.

➢ The exocytosis of the acrosomal vesicle to release its enzymes.

➢ The binding of the sperm to the extracellular envelope (vitelline layer or zona pellucida) of the egg.

➢ The passing of the sperm through this extracellular envelope.

➢ Fusion of egg and sperm cell plasma membranes.

Role of calcium in cortical granule

Upon fertilization, the intracellular calcium ion concentration of the egg increases greatly. In this high-calcium environment, the cortical granule membranes fuse with the egg plasma membrane, releasing their contents.

Once the fusion of the cortical granules begins near the point of sperm entry, a wave of cortical granule exocytosis propagates around the cortex to the opposite side of the egg. The release of calcium from intracellular storage can be monitored visually using calcium activated luminescent dyes such as aequorin (isolated from luminescent jellyfish) or fluorescent dyes such as fura-2. These dyes emit light when they bind free calcium ions.

Several experiments have demonstrated that calcium ions are directly responsible for propagating the cortical granule reaction, and that these calcium ions are stored within the egg itself.

. Structure of sperm

Each sperm consists of,

1) A haploid nucleus,

2) A propulsion system to move the nucleus, and

3) A sac of enzymes that enable the nucleus to enter the egg.

During the course of sperm maturation, the haploid nucleus becomes very streamlined, and its DNA becomes tightly compressed

In front of this compressed haploid nucleus lies the acrosomal vesicle, or acrosome, which is derived from the Golgi apparatus and contains enzymes that digest proteins and complex sugars; thus, it can be considered a modified secretory vesicle.

These stored enzymes are used to lyse the outer coverings of the egg. In many species, such as sea urchins, a region of globular actin molecules lies between the nucleus and the acrosomal vesicle.

These proteins are used to extend a fingerlike acrosomal process from the sperm during the early stages of fertilization.

. Different steps of cleavage in development of snail

Spiral holoblastic cleavage is characteristic of several animal groups, including annelid worms, some flatworms, and most molluscs. It differs from radial cleavage in numerous ways..

First, the cleavage planes are not parallel or perpendicular to the animal-vegetal axis of the egg; rather, cleavage is at oblique angles, forming a "spiral" arrangement of daughter blastomeres Second, the cells touch one another at more places than do those of radially cleaving embryos. In fact, they assume the most thermodynamically stable packing orientation, much like that of adjacent soap bubbles.

Third, spirally cleaving embryos usually undergo fewer divisions before they begin gastrulation, making it possible to follow the fate of each cell of the blastula.

Blastulae produced by radial cleavage have no blastocoel and are called stereoblastulae

The first two cleavages are nearly meridional, producing four large macromeres (labeled A, B, C, and D). In each successive cleavage, each macromere buds off a small micromere at its animal pole

. Neurlation defects

Different neural tube defects are caused when various parts of the neural tube fail to close.

Failure to close the human posterior neural tube regions at day 27 (or the subsequent rupture of the posterior neuropore shortly thereafter) results in a condition called spina bifida, the severity of which depends on how much of the spinal cord remains exposed. Failure to close the anterior neural tube regions results in a lethal condition, anencephaly. Here, the forebrain remains in contact with the amniotic fluid and subsequently degenerates. Fetal forebrain development ceases, and the vault of the skull fails to form. The failure of the entire neural tube to close over the entire body axis is called Craniorachischisis. Collectively, neural tube defects are not rare in humans, as they are seen in about 1 in every 500 live births

. Cleavage in mammals

The mammalian oocyte is released from the ovary and swept by the fimbriae into the oviduct. Fertilization occurs in the ampulla of the oviduct, a region close to the ovary.

Meiosis is completed at this time, and first cleavage begins about a day later. Cleavages in mammalian eggs are among the slowest in the animal

kingdom about 12-24 hours apart. Meanwhile, the cilia in the oviduct push the embryo toward the uterus; the first cleavages occur along this journey.

The most crucial, difference between mammalian cleavage and all other types involves the phenomenon of compaction The cells of the compacted 8-cell embryo divide to produce a 16-cell morula. The morula consists of a small group of internal cells surrounded by a larger group of external cells.

Most of the descendants of the external cells become the trophoblast (trophectoderm) cells. This group of cells produces no embryonic structures. Rather, it forms the tissue of the chorion, the embryonic portion of the placenta.

These cells generate the inner cell mass (ICM), which will give rise to the embryo and its associated yolk sac, allantois, and amnion. ***By the 64-cell stage, the inner cell mass (approximately 13 cells) and the trophoblast cells have become separate cell layers, neither contributing cells to the other group. During a process called cavitation,*** the trophoblast cells secrete fluid into the morula to create a blastocoel. The inner cell mass is

positioned on one side of the ring of trophoblast cells. The resulting structure, called the blastocyst, is another hallmark of mammalian cleavage.

Once out, the blastocyst can make direct contact with the uterus. The uterine epithelium (endometrium) "catches" the blastocyst on an extracellular matrix containing collagen, laminin, fibronectin, hyaluronic acid, and heparin sulfate receptors

What is cleavage

Cleavage is a series of extremely rapid mitotic divisions wherein the enormous volume of zygote cytoplasm is divided into numerous smaller cells. These cells are called blastomeres, and by the end of cleavage, they generally form a sphere known as a blastula.

Define isolecithal

At one extreme are the eggs of sea urchins, mammals, and snails. These eggs have sparse, equally spaced yolk and are thus isolecithal (Greek, "equal yolk")

Patterns of embryonic cleavage

Indeed, different organisms undergo cleavage in distinctly different ways. The pattern of embryonic cleavage particular to a species is determined by two major parameters:

1. ***The amount and distribution of yolk protein within the cytoplasm.***
2. ***The factors in the egg cytoplasm that influence the angle of the mitotic spindle and the timing of its formation.***
3. ***The yolk-rich pole is referred to as the vegetal pole; the yolk concentration in the animal pole is relatively***

low. The zygote nucleus is frequently displaced toward the animal pole. In general, yolk inhibits cleavage

The Fast block to polyspermy is achieved by changing the electric potential of the egg plasma membrane. This membrane provides a selective barrier between the egg cytoplasm and the outside environment, and the ionic

concentration of the egg differs greatly from that of its surroundings. This concentration difference is especially significant for sodium and potassium ions. Seawater has a particularly high sodium ion concentration, whereas the egg cytoplasm contains relatively little sodium. The reverse is the case with potassium ions.

Difference between dextral coiling and sinistral coiling.

Some snails have their coils opening to the right of their shells (dextral coiling), whereas other snails have their coils opening to the left (sinistral coiling).

Four level of gene expression

The regulation of gene expression can be accomplished at several levels:

Differential gene transcription:

It regulates that which of the nuclear genes are transcribed into RNA.

Selective nuclear RNA processing:

It regulating which of the transcribed RNAs (or which parts of such a nuclear RNA) enter into the cytoplasm to become messenger RNAs.

Selective messenger RNA translation:

It regulates that which of the mRNAs in the cytoplasm become translated into proteins.

Differential protein modification:

It regulates that which proteins are allowed to remain or function in the cell.

Explain the spina bifida

Failure to close the human posterior neural tube regions at day 27 (or the subsequent rupture of the posterior neuropore shortly thereafter) results in a condition called spina bifida, the severity of which depends on how much of the spinal cord remains exposed.

What is cone growth

Dendrites are often numerous and do not extend far from the neuronal cell body, or soma, axons may extend for several feet.The Nerve outgrowth is led by the tip of the axon, called the growth cone. The growth cone moves by the elongation and contraction of pointed filopodia called Microspikes. These microspikes contain microfilaments, which are oriented parallel to the long axis of the axon.

What is hypoblast

Area pellucida cells migrate individually into the subgerminal cavity to form the polyinvagination islands (primary hypoblast). Shortly thereafter, a sheet of cells from the posterior margin of the blastoderm (distinguished from the other regions of the margin by Koller's sickle, a local thickening) migrates anteriorly to join the polyinvagination islands, thereby forming the secondary hypoblast.

Meta cephalon

The Metencephalon gives rise to the cerebellum, the part of the brain responsible for coordinating movements, posture, and balance.

. Transcription factor

Transcription factors domains

Transcription factors have three major domains.

1) DNA-binding domain

2) Protein-protein interaction domain

3) Trans-activating domain **DNA-binding domain:**

It recognizes a particular DNA sequence.

Protein-protein interaction domain:

It allows the transcription factor's activity to be modulated by TAFs or other transcription factors.

Trans-activating domain:

It activates or suppresses the transcription of the gene whose promoter or enhancer it has bound. Usually,

this trans-activating domain enables the transcription factor to interact with proteins involved in binding RNA polymerase.

MITF:

(Microphthalmia-Associated Transcription Factor. The microphthalmia (MITF) protein is necessary for the production of pigment cells and their pigments.

Pax6:

Paired box gene 6. The Pax6 transcription factor, which is needed for mammalian eye, nervous system, and pancreas development, contains two potential DNA-binding domains.

Pdx1:

Pancreatic And Duodenal Homeobox 1 gene. It is specific for the pancreatic region of the endoderm.

Cortice and nuclei

In the cerebellum, some neuronal precursors enter the marginal zone to form clusters of neurons called nuclei

Each nucleus works as a functional unit, serving as a relay station between the outer layers (cortices) of the cerebellum and other parts of the brain

Types of cleavage in snail

At one extreme are the eggs of sea urchins, mammals, and snails. These eggs have sparse, equally spaced yolk and are thus isolecithal (Greek, "equal yolk"). In these species, cleavage is holoblastic (Greek holos,

"complete"). meaning that the cleavage furrow extends through the entire egg.

Zygotes containing large accumulations of yolk undergo meroblastic cleavage, wherein only a portion of the cytoplasm is cleaved. The cleavage furrow does not penetrate into the yolky portion of the cytoplasm.

The eggs of insects have their yolk in the center (i.e., they are centrolecithal), and the divisions of the cytoplasm occur only in the rim of cytoplasm around the periphery of the cell (i.e., superficial cleavage).

The eggs of birds and fishes have only one small area of the egg that is free of yolk (telolecithal eggs), and therefore, the cell divisions occur only in this small disc of cytoplasm, giving rise to the discoidal pattern of cleavage.

Events in primary neurulation

During primary neurulation, the original ectoderm is divided into three sets of cells.

- The internally positioned neural tube, which will form the brain and spinal cord.
- The externally positioned epidermis of the skin.

The neural crest cells. The neural crest cells form in the region that connects the neural tube and epidermis, but then migrate elsewhere; they will generate the peripheral neurons and glia, the pigment cells of the skin, and several other cell types

What type of molecules involved in histone protein.

Nucleosome is composed of an octamer of histone proteins (two molecules each of histones H2A-H2B and histones H3-H4) wrapped with two loops containing approximately 140 base pairs of DNA.

Difference b/w Karyokinesis and Cytokinesis

karyokinesis the mitotic division of the nucleus.

The mechanical agent of this division is the mitotic spindle, with its microtubules composed of tubulin (the same type of protein that makes up the sperm flagellum).

The second process is cytokinesis the division of the cell.

The mechanical agent of cytokinesis is a contractile ring of microfilaments made of actin (the same type of protein that extends the egg microvilli and the sperm

acrosomal process).

The relationship and coordination between these two systems during cleavage is depicted in Figure, in which a sea urchin egg is shown undergoing first cleavage.

meroblastic

The cell divisions do not completely divide the egg, so this type of cleavage is called meroblastic.

.Does ascidians have notochord

Ascidians, members of the tunicate subphylum, are fascinating animals for several reasons, but the foremost is that they are invertebrate chordates. They have a notochord as larvae (and therefore are chordates), but they lack vertebrae.

What BHLH protein specific for.

the myogenic bHLH proteins are specific for muscle cells. Any cell making a myogenic bHLH transcription factor such as MyoD or Myf5 is committed to becoming a muscle cell.

.What is promoter

Promoter:

A region, which is responsible for the binding of RNA polymerase and for the subsequent initiation of transcription.

Define air space

An air space forms when the contents of the egg cool and contract after the egg is laid. The air cell usually rests between the outer and inner membranes at the egg's larger end.

Describe osteoclast in detail

Osteoclasts are probably derived from the same precursors as macrophage blood cells, and they dissolve both the inorganic and the protein portions of the bone matrix.

Each osteoclast extends numerous cellular processes into the matrix and pumps out hydrogen ions onto the surrounding material, thereby acidifying and solubilizing it.

The blood vessels also import the blood-forming cells that will reside in the marrow for the duration of the organism's life. The number and activity of osteoclasts must be tightly regulated.

If there are too many active osteoclasts, too much bone will be dissolved, and osteoporosis will result.

Conversely, if not enough osteoblasts are produced, the bones are not hollowed out for the marrow, and osteopetrosis results.

. Write classes of cadherins

In vertebrate embryos, several major cadherin classes have been identified:

1. E-cadherin

2. P-cadherin

3. N-cadherin

4. Protocadherins

E-cadherin: a. Epithelial cadherin is expressed on all early mammalian embryonic cells, even at the 1cell stage. Later, this molecule is restricted to epithelial tissues of embryos and adults.

P-cadherin: a. Placental cadherin appears to be expressed primarily on the trophoblast cells (those placental

cells of the mammalian embryo that contact the uterine wall) and on the uterine wall epithelium.

N-cadherin: a. Neural cadherin is first seen on mesodermal cells in the gastrulating embryo as they lose their

E-cadherin expression. It is also highly expressed on the cells of the developing central nervous system.

Protocadherins: a. Calcium-dependent adhesion proteins that differ from the classic Cadherins in that they

lack connections to the cytoskeleton through catenins. Protocadherins have been found to be very important in separating the notochord from the other mesodermal tissues during Xenopus gastrulation.

Definer silencer

Enhancers can also be used to inhibit transcription. In some cases, the same transcription factors that activate the transcription of one gene can be used to repress the transcription of other genes. These

"negative enhancers " are also called silencers.

Difference between sex and reproduction

Sex and reproduction are two distinct and separable processes. Reproduction involves the creation of new individuals. Sex involves the combining of genes from two different individuals into new arrangements.

Sex and reproduction are two different processes that can or not to occur simultaneously. Sex must be considered as an exchange of genetic material doing new combinations, that can originate or not new individuals. While that the reproduction has that obligatorily to originate new beings.

Differentiation process of medulla and spinal chord

As the cells adjacent to the lumen continue to divide, the migrating cells form a second layer around the original neural tube. This layer becomes progressively thicker as more cells are added to it from the germinal neuroepithelium. This new layer is called the mantle (or

intermediate) zone, and the germinal epithelium is now called the ventricular zone (and, later, the ependyma).

The mantle zone cells differentiate into both neurons and glia. The neurons make connections among themselves and send forth axons away from the lumen, thereby creating a cell-poor marginal zone.

Eventually, glial cells cover many of the axons in the marginal zone in myelin sheaths, giving them a whitish appearance. Hence, the mantle zone, containing the neuronal cell bodies, is often referred to as the Gray matter; the axonal, marginal layer is often called the White matter. In the spinal cord and medulla, this

basic three-zone pattern of ependymal, mantle, and marginal layers is retained throughout development.

The gray matter (mantle) gradually becomes a butterfly-shaped structure surrounded by white matter; and both become encased in connective tissue.

As the neural tube matures, a longitudinal groove the sulcus limitans divides it into dorsal and ventral halves. The dorsal portion receives input from sensory neurons, whereas the ventral portion is involved in effecting various motor functions.

1. The first cleavage furrow appears centrally in the_____ **blastodisc**
2. . Fertilization occurs in the ***ampulla*** of the_____ **oviduct**
3. . ***Mother*** provides fetus with_____ ***nutrients and oxygen***
4. . The first segregation of cells within the inner cell mass results in the formation of the _____ ***hypoblast (sometimes called the primitive endodem)***
5. The fetal portion of placenta formed by_____***Chorion***
6. . Paracrine factor instruct ***myotome*** cell to become_____ ***Muscles***
7. . When ***myoblasts*** stop dividing, secrete _____***fibronectin***

8. . ***Retinoic Acid*** comes from _____***Somites***
9. . ***Endochondral ossification*** involves the formation of ___ cartilage
10. ***J.Hammerling*** exchanged nuclei b/w two ***morphologically*** distinct species
11. ***Promoter is enhancer*** which is thought to _____***regulate transcription***
12. Identical twins share a common_____ **_Chorion**
13. Which regulate function of Promoter_____ **Enhancer**
14. . The space between two neurons, where transmitter are released synapse integral components of the neural tube lining become_____ **ependymal**
15. . Mantle cells differentiate into ____***Gray matter***
16. . The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the _____ **area_opaca.**
17. . During fertilization, the ***actin molecules*** fuse to form the long _____***microfilaments***
18. . A condition in which cell lack the instructions for development____ **naive**
19. metamorphosis is initiated by hormones from the tadpole's _____**thyroid gland**
20. . Brain has __ **100 billion neurons**
21. . Release of Ca from intracellular storage controlled by _dye_____**Aequorin**
22. . Nervous system develop by _____”____**Neural Tube**
23. . Allows the mRNA to exit the nucleus_____ ____**poly a tail**
24. Mode of commitment_____ 3
25. strageries of commitment_____2
26. ***Rana pipiens usually lays around 2500 eggs*** _while the bullfrog, Rana catesbiana, can lay ***many as 20,000***
27. Multinucleated type of cell form.__ ____ _ **syncytiotrophoblast**
28. slow block, initiated by the intracellular release of _____ **calcium ions**
29. There are ***<u>two</u>*** major modes of bone formation, or ***osteogenesis***
30. Mutations of the ***<u>SOX9 gene</u>*** _cause ***camptomelic dysplasia***, a rare disorder of skeletal development that results in deformities of most of the bones of the body
31. Human brain contains_____ **10" glial cells**

32. Telolecithal is cleavage is found in eggs of _____**Birds and Fishes**

33. ***Pax6***. plays a major role in _____ **eye formation**

34. ***Pax1 _and Scleraxis*** are the transcription factors are thought to ***activate cartilage specific genes***

35. The eggs of insects have their yolk in the center (i.e., **they are centrolecithal).**

36. Egg activates due to which ion more_____**Ca**

37. Which dermis give rise to brain_____ **Ectoderm**

38. Which dermis give rise to eye _____***Ectoderm***

39. Mod of commitment in insects syncytial specification

40. Regulative development arises from conditional specification

41. Nucleosome is composed of an octamer of histone proteins wrapped with two loops containing approximately _____***140 base pairs_of DNA***

42. the ***membrane potential of the sperm becomes more negative*** as potassium ions leave the sperm

43. Paracrine factors such as ***TGF-and FGF7 a***re important in normal skin development

44. In mouse ***ZP3*** is attached to ***zona pellucid***

45. It is estimate Legg is ***25000 to 50000 mRNA.*** _____ **Sea Urchin**

46. ***Fertilin*** protein is important for spermegg fusion in ___ _ ***Mammals***

47. Fast block to polyspermy is initiated by __ Sodium

48. Transition from fertilization to cleavage is caused by factor____ **MPE**

49. Eggs having sparse and equally yolk space are called _____***isolethial***

50. Archenteron is used for _____**primitive gut**

51. ***Spiral cleavage found in Snails***

52. ***Sonic hedgehog is a type of protein***

53. Which type of neurulation in fish____ **secondary**

54. Myogenic HLH proteins are specific for development of _____ ***muscle cell***

55. ***Zygote*** which divide mitotically to produce all the cells of the body.

56. ***Tatratology*** the study of birth defect.

57. Identical twins occur in roughly ***0.25% _of human births.***

58. Requlatively development is seen in most vertebrate embroys, and it is obviously ***Critical developmemt of identical twins.***

59. Insect such as_***drosophila_use all three mods of specification*** to commit their cells to particular fates.

60. The speed of metamorphosis is ***directly linked with environment pressures***

61. In acetabularia unicellular organisms, development us controlled ***at both the transciptional and translational level.***

62. The individuals of each species, however is divided into ***two mating types_Plus and minus calamadomonas***

63. ***140 base pairs of DNA _____ nucleosome.***

64. ***Viteline*** envelop contains at least eight different ___ ***glycoproteins***.

65. There are ***two major modes of bone formation,*** or osteogenesis

66. ***Telolecithal*** is cleavage is found in eggs of ***Birds and Fishes***

67. The eggs of insects have their yolk in the center (i.e., ***they are centrolecithal***).

68. Egg activates due to which ion more____-**Ca**

69. Which dermis give rise to brain _____***Ectoderm***

70. Which dermis give rise to eye _____ ***Ectoderm***

71. Mod of commitment in insects_____ ***syncytia! specification***

72. Regulative development arises from ____ ***conditional specification***

73. Nucleosome is composed of an octamer of histone proteins wrapped with two loops containing approximately ___ ***140 base pairs of DNA***

74. the membrane potential of the sperm becomes negative as potassium ions leave the _____ ***zona sperm***

75. **Paracrine factors** such as TGF-and FGF7 are important in normal **skin development**

76. Fast block to polyspermy is initiated by_____ **Sodium**

77. Transition from fertilzation to cleavage is caused by factor_ ____**MPF**

78. Eggs having sparse and equally yolk space are called _____**isolathial**

79. Archenteron is used for_____ ***Primitive gut***

80. Spiral cleavage found in _____ . **Snails**

81. Sonic hedgehog is a type of-_____ **protein**

82. Nural crest divides functionally in domains _____4

83. In acetabularia unicellular organisms, development us controlled at both the transciptional.and translational level

84. The individuals of each species, however is divided into two mating types Plus and minus ____ ***Chlamydomonas***

85. The first evidence suggestion the importance of sperm in reproduction came from a series experiment performed by___ ***Lazzaro Spalanzani***

86. The volume of the sea urchin is abou t ***200Pico litres***

87. The embryo contains three gem layers The ***ectoderm,the endoderm and the mesoderm.***

88. ***The neural form keel fish***

89. ***Spiral holoblastic clevage is characteristic of several animals groups, including annelid worms, some Tlatworms, and most mollusks (snails).***

90. Diploblastic animals are those who have ectoderm and endoderm but no true ***mesoderm*** .

91. Even distribution of yolk in Ovum_____-***isolecithal***

92. After the blastoderm cells have covered about half the zebra fish yolk cell, a ***thickening***

93. occurs throughout the margin of thee epibolizing blastoderm. This thicken ingis called _____**germ ring**

94. Which forms spinal cord, hind brain and mid brain,_chordamesoderm

95. This streak is first visible in birds as a thickening of the epiblast at the posterior region

96. Of the embryo , Just anterior to ***koller sickle.***

97. It will nduce the uterine cells of the mammals to from the matrenal portion of the placenta, the_____***Decidua***.

98. Between the area pellucida and the area opaca is a thin layer of cells called the _____***marginal zone (or marginal belt).***

99. The mitotic division of the nucleus is called Karyokinesis

100. Fishes,reptiles and birds undergo ***discoidal meroblastic cleavage***, wherein the early cell

101. divisions do not cut through the yoIk or the exit

102. The ***fast block, initiated by sodium influx*** into the cell in **sea urchin.**

103. The cells adjacent to the chordamesoderm, the paraxial mesoderm cels, are the precursors of the mesoderma somites in the fish.

104. The rudiment of the central nervous system, is called neurulation.

105. Blastula produced by radial cleavage has no blastocoels and are called ***stereoblastulae***

106. Fraternal twins are the result of __ ***two separate fertilization events.***

107. The prechordal plate induces the formation of formation of the midbrain, hindbrain, and spinal cord.

108. The resulting larvae of snail is called glochidia which resemble tiny bear traps.

109. Fertilization in the absence of sex occur through the_____***Binary Fission***

110. The **macromeres** divide meridionally, forminga tier of eight (8)

111. cells, derived from the micromeres, are called the primary mesenchyme

112. Molluscus follow which type of cleavage spiral holoblastic cleavage

113. A mammalian bl***astula is called a blastocyst.ells***

114. Which of the following undergo discoidal cleavage fishes, reptiles and birds

115. The egg jelly factors that initiate the Acrosomal reaction n sea urching are often ***highly specific to each species***

116. The Mesencephalon gives rise to the ***cerebellumn***

117. In primary **neurulation**, the original ectoderm is divided into ***three sets of cells***

118. The growing region at the tip of the lip is called the ***chordoneural hinge***

119. Neural Tube will form ***Brain and spinal cord***

120. Each ICM (Ihner Cell Mass) cell has the same_____.**potency**

121. Variable cleavages produce ***no invariant fate*** assignments to cells

122. In early embryos of insects cell division is ***not complete***

123. The neural crest cells originate at the dorsal most region of the ***neural tube***

124. The human globin gene has an enhancer in its 3 UTR, roughly ***700 base pairs***

125. the 5ox9 gene which encodes a DNAbinding protein

126. Osteoclasts are probably derived from the same precursors as macrophage blood cells

127. The direct conversion of ***mesenchymal tissue into bone is called intramembranous ossification***

128. Those cells that remain integral components of the neural tube ining become ***Ependymal cells***

129. The original type of trophoblast cells constitute a layer called the ***cytotrophoblast***

The major structural characteristic of avian, reptilian, & mammalian gastrulationis _____ ***primitive streak***

130. Both hae ***Anurans*** and ***Toad and fros***

131. They have a notochord as larvae (chordates), but they ***lack vertebrate***

132. Acrosomal vesicles is dertved from Golgi apparatus and qnta ins Enzymes The large subunit of MPFI5____ ***cyclin B***

133. . ZPS is the specitic ghycoprotein in the **mouse**

134. . MITF necessary for the production of_**Pigment cells**

135. Mutation of sox9 gene causecamptomelic **dysplasia**

136. Between the blastoderm and the yolk is a space called the sub-germinal cavity The second step of gastrulation involves the involution of the **mesoderm**

137. The volume of the sea urchin is about_***200 Pico liters***

138. Hindbrain Rhombencephalon

139. The most crucial, difference between mammalian cleavage and all other types involves the **phenomenon of compaction**

140. The embryo contains three germ layers. The ectoderm, the endoderm and the mesoderm.

141. Diploblastic animals are those who have ectoderm and endoderm but no true **mesoderm (hydra)**

142. Cell fate is determined by signaling. The micromeres constitute a major signalling center.

143. B-catenin is important for the inducing capacity of ***micromeres***.
144. Size of bindin protein ___**30,500-Da**
145. which has two parts in cell cycle options _____***Blastomere***
146. After the blastoderm cells have covered about half the zebra fish yolk cell, a thickening occurs throughout the margin of the epibolizing blastoderm. This thickening is **called the germ ring**
147. Which forms spinal cord, hind brain and mid brain__**chordamesoderm**.
148. This ***streak*** is first visible in birds as a thickening of the Epiblast at the posterior region of the embryo, just anterior to ***koller sickle.***
149. Fishes, reptiles, and birds undergo ***discoidal meroblastic cleavage, w***herein the early cell divisions do not cut through the yolk of the egg.
150. The fast block, initiated by ***sodium influx i***nto the cell in **sea urchin.**
151. The cells adjacent to the chordamesoderm, the paraxial mesoderm cells, are the precursors of the ***mesodermal somites in the fish.***
152. The rudiment of the central nervous system, is called ***neurulation***.
153. Blastula produced by radial cleavage has no blastocoel and are called ***stereoblastulae***.
154. Fraternal twins are the result of ***two separate fertilization events.***
155. The prechordal_plate_induces the formation of the forebrain; the **chorda mesoderm induces** The formation of the midbrain, hindbrain, and **spinal cord.**
156. Due to Loss in hydrogen ions, ***pH rises***.
157. As a result of ***gastrulation***, ectoderm, mesoderm and Endoderm are formed
158. Male leopard Frogs make their sperm ***in summer.***
159. ***Differential gene transcription is step for gene expression regulation***
160. Coding regions of DNA_____ ***exon***.
161. How many domains are present for transcription ____**3**
162. pdx1 is present in **Pancreatic And Duodenali gene.**
163. Bacteria transmit genes from one cell to other by **sex pili.**
164. Cells rearrange their DNA to create new **immunoglobin an antigen receptor gene.**
165. Sea urchins contain **25,000 to 50,000 mRNA.**
166. In many species, sperm are attracted towards eggs by ____***chemotaxis***

167. Acrosomal protein mediating recognition by protein **bindin**
168. Sea water has High **sodium ion conc.**
169. Eggs have sparse equally spaced yolk are **icolecithel**.
170. Cytoplasm of blastodisc become embryo this meroblasticn division called ***discoidal***.
171. The micromere constitute major **signaling center.**
172. Osteoblasts secrete collagen proteoglycan matrix that is able to bind ***calcium salts***
173. Tgf beta family secreted from the surface ectoderm and ***roof of neural tube.***
174. Dendrites are numerous and do not extend far from the neuronal cell body or soma, axons may extend for several Join **WUBE feet.**
175. One agent of ventral specification is ***sonic hedgegog protein***
176. In frogs and chicks, secondary nueralation is seen in ***neural tube.***
177. The cells at dorsal most portion of neural tube become ***neural crest cells.***
178. About 33% of identical twins have two complete and separate ***chorions***
179. Mammals undergo ***holoblastic rotational cleavage***
180. In birds, gravity is critical in determining anterior posterior axis
181. Decidua becomes rich in blood vessels that will provide oxygen and nutrients to ***embryo***
182. Mammalian **oocyte** is released from ovary and swept by fimbriae into the **oviduct**,
183. Fertilization in chick egg occurs in oviduct.
184. Developmental decisions are usually_____**irreversible**
185. All multicellular organisms arise by progressive change called ***development***..
186. Major class of cadherin includes ***N-cadherin***.
187. Cortical are granules are membrane bound structure containing ***proteolytic enzymes.***
188. Small subunits of **MPF cyclin-dependent kinase.**
189. It secretes a cellulose tunic (which gave the name "tunicates" to these creatures).
190. Inner surface cells thin____ ***30x5um process filopodia.***
191. ***B-catenin*** is Important for Inducing capacity of ***micromeres***.

192. In fishes, ***First 12 divisions occur synchronously*** form a mouin of cells that sits an animal] ***pole of large yolk cell.***

193. Fertilization occurs in ***ampulla of oviduct a close, a region close to ovary.***

194. In amphibians most of the xenopus most of tadpole is formed by primary neurulation and ***tail of neural tube formed by secondary neuralation.***

195. ***The marginal zones to form clusters of neurons called nuclei***.

196. Paracrine factors instruct Myotome cells to become muscles by inducing them to synthesize ***MyoD protein.***

197. Sclerix is expressed in mesenchyme from the ***scleretome***.

198. ***Acrosomal*** reaction in sea urchins is often highly specific to each species.

199. Cells are arrange their.. to creat new ***immunoglobin***

200. Regulation of gene expression by ***several level.***

201. Which type of neuralation in fish ***exclusively secondary***

202. sonic hedgehocy is a type of ***protein***, .

203. How many strategies involve in cell commitment __ ***2.***

204. Resting membrane potential of the egg is_____ ***70mv***.

205. One chemotactic molecule, a 14-amino acid peptide called resact

206. . Force for sperm propulsion is provided by dynin.

207. . Regulative development arises from ***Conditional Specification***

208. Fish have discoidal ***telolicithal*** pattern.

209. The mitotic spindle and contractile ring are ***perpendicular*** to each other.

210. Gastrulation in Snails is accomplished primarily ***by epiboly.***

211. The first cell movement of fish gastrulation is the epiboly of the ***blastoderm***.

212. This fetal organ the Chorion is derived primarily from embryonic ***trophoblast*** cells.

213. . ***Secondary neurulation*** involves the making of a medullary cord and its subsequent hollowing into neural tube

214. ***The mantle zone cells differentiate into both neurons and glia.***

215. The cerebral cortex in humans has ***six layers.***

216. Paracrine factors such as ***TGF and FGF7*** are important in normal skin development

217. . The fifth phase involves the invasion of the cartilage model by blood vessels.

218. **Rana pipiens usually lays around 2500 eggs** while the bullfrog, ***Rana catesbiana, can lay as many as 20,000***

219. Multinucleated type of cell form___ . ***syncytiotrophoblast***

220. **slow block**, initiated by the intracellular ***release of calcium ions***

221. ***In the sea urchin all region of the egg plasma membrane are capable of fusing with sperm .*** In the several other species certain region of the membrane are specially f***or sperm recognization and fusion .***

222. Fusion is an active process often mediated by specific ***fusogenic protein?_____True***

223. The egg of birds and fish have only one small area of the egg that is free of yolk called _____**Telolecithal**

224. when added to a suspension of Arbacia sperm _____ **Resact**

225. Placing unfertilized sea urchin egg into sea water containing _ cause the cortical granules reaction and the cleavage of the fertilization envelope_____ **A23187**

226. Zygote containing large accumulation of yolk undergo _ cleavage where in only a portion of the cytoplasm is cleaved._____**Mesoblastic**

227. ***The second part of the acrosomal reaction involve the extension of the acrosomal process . This profusion arises from the polymerization of the globular into actin filament__True***

The contractile ring create a cleavage furrowe which eventually bisect the plan of mitosis thereby treating two genetically equalent blatomere_____ True

228. Miler found that when sperm were added to oocyte that had not yet complete their _ ***mitotic division .***

229. The drug _ is a calcium ionophore_____ ***A23187***

230. The sea urchin has _ mechanism to avoid polyspermy?_____**2**

231. A cytoplasm that contains may include is called a _ ._____**Cyntium**

232. The controlling agent of the efficiency and rate of transcription is _ ___ **Enhancer**

233. Cytoplasm determinant are the molecule in _ that can be help to determine cell fate_____ ***Cytoplasm***

A. Leeuwenhoek the Dutch microscopic who discovere sperm in _____***1678***

234. _ produced by radial cleavage have no blatocoel and are called stereo blastula.__. ***Blastula***

235. The opening of the archeteron at the ugetal region is called the_____ **Blastopore**

236. The technique which is onsite the hybridization is _____**PCR**

237. In the most Marin invertebrates the acrosomal reaction has _ component _____**1**

238. Autonomous specification give rise to a pattern of development reffed to as_____***Mosaic development***

239. Using imological technique _ demonstrate that bindin is located specifically on the ***acrosomal process***

240. The division of the cytoplasm occurs in the cytoplasm around the periphery of the cell process called as_____**Superficial cleavage**

241. The invage inated region formulation during gastrulation in sea urchin is called as_____ ***Blatocoel***

242. The cortical granules are membrane bound structure containing _ enzyme____ ***Proteolytic***

243. The fast sperm inhibition condition maintained by the plasma membrane which stead fastly inhibit the entry fusion into the environment_____ ***Sodium***

244. In the drosophila interior most portion of the egg contain an mRNA that encode a protein is called _._____ ***Bicoid***

245. The cell division occurs only in small disc of cytoplasm giving rise to the pattern of cleavage+_____ ***Discoidal***

246. Transcription factor have _ major domain._____***3***

247. Cycling B regulate the small subunits of the cyclin dependent kinase_____ ***MPF***

248. RNA polymerase bind with the _ in the start of transcription _____. ***Promotor***

249. The eggs of the insects have their yolk in the center _____. ***Centrolecithal***

250. In frogs another ovarian hormone progesterone signal the egg to resume it's division_____**Mitotic**

251. _ is the entire range of the cell type a particular cell can give rise to in all possible environment_____. ***Potency***

252. _ in different species are those organ whose similarly due to sharing their common ancestral structure_____ ***Homologous structure***

253. red blood cells makes .....*****globins***____

254. once the three germs layers are established , the cell interact with one another and rearrange themselves to produce .._____***organogenesis***____

255. The ability of the embryonic cells to change their fates to compensate for the missing part is called ……_____***regulation***_

256. ?The zygote nucleus is frequently displayed toward the animals pole . in general ……inhibits cleavage _____**yolk**

257. The asymmetric distribution of cytoplasmic determinants indicates that the mechanism of differentiation is entirely…_____***intinsic***

258. several experiments have demonstrated that calcium ions are ……. ***Responsible for propagating the cortical .***

259. . ***The mechanism of chemotaxis differ among species***

260. Cell fate is equal to potency plus ……. _____***environmental***

261. In evolving sexual reproduction ……..important advances had to be achieved ?/_____***2***

262. Finally a fourth cortical granule protein ……..forms a coating around the egg_____**Mucopolysaccharides**

263. The negaytive enhancers are called …….._____ ***silencers***

264. The sperm and egg plasma membrane then join together and material from the (….) is widened by the ……?_____ **Actin , polymerization**

265. The identical cells can follow ……. if one is expressed to an external signal (often secreted by a different cells)_____ ***Alternative fates***

266. The mitotic division of the nucleus is called ……._____ ***karyolinesis***

267. Directly beneath the sea urchin ewgg plasma membrane are about ……. Cortical granules ?_____ ***1500***

268. …… is the study of how anatomy changes during the development of different organism ._____***comparative anatomy***

269. eukaryotic genes are contained with a complex of DNA and protein called ._____**chromatin**

270. The _ intiated but sodium influx into the cell in sea urchin ____ ***Fast block***

271. The release of calcium from intracellular storage can be monitored visually using calcium-activated luminescent dyes such as _____***Fura***

272. The shell is the egg outer covering accounting for about **9 to 12% of its weight** depending upon egg size

273. the shell is first line of defense against bacterial contamination this bumpy and gainy outer covering has approximately _ tiny pores and is made almost entirely of calcium carbonate?____**1700**

274. If we insert a electron into an egg and placed a second electrode to it we can measure the constant difference in charge across the egg.____ ***Plasma membrane***

275. Sex without reproduction is also common among _ organism ___ ***Unicellular***

276. The _ is the hierarchical series of decesion involving specific pre.existing cell type_____ ***Development hierchery***

277. In frog estrogen is a hormone that can instruck the _ to make and secret the blood in to the enlarge the cell egg in the ovary_____***Liver***

278. The acrosomal protein mediating this recognition is called ***bindin***

279. Cyclic activity of MPF is actually the result of two coordinated process_____***Cytokinesis***

280. When the membrane are fused the sperm nucleus mitochondria centrioles and flagellum can enter the egg?_____ ***True***

281. _ produced by radial cleavage have no blatocoel and are called stereo blastula___***Blastula***

282. Spiral holoblatic cleavage is the characteristics several animals group like__ ***Molluscus***

283. The Wilcox and colleagues _ found that nearly all human pregnancies results from sexual intercourse during a six day period ending on the day of evolution ?_____ **1995**

284. Shortly after the calcium ion level rise in a sea urchin egg it's ***intracellular pH also increase***

285. the rise in the intracellular pH begin with a second influx sodium ion which causes a 1:1 exchange between the ion from the sea water and hydrogen ion from the egg?__True

286. The yolk rich pole is reffed to as the _ ._____ **Vegetal pole**

287. The egg microvilli attache to _ provide support for the blastomere _____ ***Hyaline laye***r

288. The individual of each species however are divided into a mating types plus and minus when a plus and minus they join their cytoplasm and their nuclei fuse to form a diploid zygote?_____True

289. The ability of the embryonic cell to change their fates to compensate for the missing part called _ _____ Regulation

ZOO501 FINAL TERM MCQS

1. Mode of commitment_____3___
2. strageries of commitment_____2___
3. Rana pipiens usually lays around __2500 eggs____while the bullfrog, Rana catesbiana, can lay as many as 20,000
4. Multinucleated type of cell form._____syncytiotrophoblast
5. slow block, initiated by the intracellular release of_____ calcium ions
6. There are_____ two ____ major modes of bone formation, or **osteogenesis**
7. Mutations of the ___*SOX9* gene____cause camptomelic dysplasia, a rare disorder of skeletal development that results in deformities of most of the bones of the body
8. Human brain contains_____**10^{12}** glial cells____
9. Telolecithal is cleavage is found in eggs of _____Birds and Fishes
10. ____Pax6 _____ plays a major role in eye formation
11. ___Pax1_____and Scleraxis are the transcription factors are thought to activate cartilage-specific genes
12. The eggs of insects have their yolk in the ____center_____ (i.e., they are centrolecithal),
13. Egg activates due to which ion more _____Ca____
14. Which dermis give rise to brain _____Ectoderm____
15. Which dermis give rise to eye _____Ectoderm_____
16. Mod of commitment in insects _____syncytial specification____
17. Regulative development arises from_____ conditional specification
18. Nucleosome is composed of an octamer of histone proteins wrapped with two loops containing approximately ___140 base pairs___ of DNA
19. the membrane potential of the sperm becomes ___more negative___ as potassium ions leave the sperm
20. Paracrine factors such as TGF- and FG**F7** are important in normal _____skin development
21. The male leopard ___frog___release sperm in the summer
22. In mouse ZP3 is attached to_____zona pellucid
23. It is estimate ____egg is 25000 to 50000 mRNA. _____Sea Urchin
24. Fertilin protein is important for sperm-egg fusion in _____Mammals___
25. Fast block to polyspermy is initiated by _____Sodium___
26. Transition from fertilization to cleavage is caused by factor _____MPF
27. Eggs having sparse and equally yolk space are called _____isolethial___
28. Archenteron is used for _____primitive gut___
29. Spiral cleavage found in _____Snails____
30. Sonic hedgehog is a type of _____protein____
31. Which type of neurulation in fish _____secondary___
32. Nural crest divides functionally in domains _____4_____
33. Myogenic bHLH proteins are specific for development of _____muscle cell___

34. Zygote which divide mitotically to produce all the cells of the body.
35. Tatratology the study of birth defect.
36. Identical twins occur in roughly 0.25% of human births.
37. Requlatively development is seen in most vertebrate embroys , and it is obviously critical in development of identical twins.
38. Insect such as drosophila use all three mods of specification to commit their cells to particular fates.
39. The speed of metamorphosis is directly linked with environment pressures
40. In acetabularia unicellular organisms, development us controlled at both the transciptional and translational level.
41. The individuals of each species, however is divided into two mating types Plus and minus calamadomonas.
42. 140 base pairs of DNA nucleosome.
43. Viteline envelop contains at least eight different glycoproteins.
44. The first evidence suggestion the importance of sperm in reproduction came from a series experiment performed by Lazzaro Spallanzani
45. The volume of the sea urchin is about 200 Pico liters
46. Hindbrain Rhombencephalon
47. The most crucial, difference between mammalian cleavage and all other types involves the phenomenon of compaction
48. The embryo contains three germ layers. The ectoderm, the endoderm and the mesoderm.
49. The neural form keel fish
50. Spiral holoblastic clevage is characteristic of several animals groups, including annelid worms, some flateworms, and most mollusks(snails).
51. Diploblastic animals are those who have ectoderm and endoderm but no true mesoderm (hydra)
52. Cell fate is determined by signalling. The micromeres constitute a major signalling center. B-catenin is important for the inducing capacity of micromeres.
53. Even distribution of yolk in ovum isolecithal
54. Size of bindin protein 30,500-Da
55. which has two parts in cell cycle options Blastomere
56. After the blastoderm cells have covered about half the zebra fish yolk cell, a thickening occurs throughout the margin of the epibolizing blastoderm. This thickening is called the **germ ring**
57. Which forms spinal cord, hind brain and mid brain chordamesoderm.
58. This streak is first visible in birds as a thickening of the Epiblast at the posterior region of the embryo, just anterior to koller sickle.
59. I will induce the uterine cells of the mammals to from the matrenal portion of the placenta, the Decidua.
60. Between the area pellucida and the area opaca is a thin layer of cells called the **marginal zone** (or marginal belt).
61. The mitotic division of the nucleus is called karyokinesis
62. Fishes, reptiles , and birds undergo discoidal meroblastic cleavage, wherein the early cell divisions do not cut through the yolk of the egg.
63. The fast block, initiated by sodium influx into the cell in sea urchin.

64. The cells adjacent to the chordamesoderm, the paraxial mesoderm cells, are the precursors of the mesodermal somites in the fish.

65. The rudiment of the central nervous system, is called neurulation.

66. Blastula produced by radial cleavage has no blastocoel and are called stereoblastulae.

67. Fraternal twins are the result of two separate fertilization events.

68. The prechordal ____plate_____induces the formation of the forebrain; the chordamesoderm induces the formation of the midbrain, hindbrain, and spinal cord.

69. The resulting larvae of snail is called _____glochidia____which resemble tiny bear traps.

ZOO501 CURRENT MCQS MIDTERM (2018) UPDATED

1. Red blood cells make_____Globins
2. how many strategies for developmental commitment _____2
3. Ribosomes produce in sea urchin _____10^{12}
4. Resact is isolated from egg jelly of _____sea urchin
5. ___is a discipline that studies embryonic & other developmental processes___Developmental Biology
6. 0nly___small___ %age of genome expressed in particular cell
7. Digestive & respiratory system formed by_____Endoderm
8. Progesterone work for signals the egg to resume its _____meiotic division
9. Insect characteristics are ..... specification_____syncytial
10. In Chlamydomonas haploid chromosome has.....copy _____one
11. Modes of commitments are..... _____3
12. After fertilization developmental process proceed by _____Cleavage
13. Ca ion responsible for cortical granules stores in _____Endoplasmic Reticulum of Eggs
14. Seawater has a particularly high ___ ion concentration _____Na (Sodium)
15. There are ___approaches of anatomy_____4(four)
16. The major cell adhesion molecules appear to be the Cadherins.
17. It is estimate ____egg is 25000 to 50000 mRNA. _____*Sea Urchin*
18. Fusion of ____and sperm cell plasma membrane. _____Egg
19. Resact diffuse readily in _____sea water.
20. The requirement of capacitance varies from species to _____Species.
21. The division of embryo into 3 _____Germ layers
22. Cell potency___≥___fate
23. Zona pellucida___ also initiate the acrosomal reaction after sperm bound to it.
24. Mammalian egg is surrounded by layer called. _____Cumulus
25. The volume of the sea urchin is about ___200___ Picoliters.
26. The sperm travel by the amoeboid motion of ____lamellipodial extensions __of the cell membrane.
27. The sex and __Reproduction___ process included into fertilization.
28. The term spermatozoa means.. _____Sperm Animals
29. In 1824____J. L. Prevost and J. B. Dumasclaimed_ sperm is not parasite.

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30. 4 cadherin classes are found in Vertebrates embryos. .
31. The second major group of protostomes are the Lophotrochozoa
32. Sperm discovered in 1678
33. Transcription factors have three major domains
34. How many Amphibian oocytes produces ribosomes as 10^{12}
35. Enhancers can also be used to inhibit transcription
36. In deutrosomes the mouth opening would formed after... opening Anal
37. Germs cell formed in.... specialized portion of egg cytoplasm
38. Syncytial specification found in insects
39. Third anatomical approach to developmental biology is Teratology
40. The mature sea urchin is metabolically sluggish and activated by. Sperm
41. The resting membrane potential is.....mv. 70mV
42. Fusion is a Active process that is activated by fusogenic proteins.
43. Axoneme s formed by. Microtubules
44. Acrosome can be modified secreted by..... Golgi apparatus (not sure)
45. The male leopard frog release sperm in the summer.
46. Eukaryotic genes are contained within a complex of DNA and protein called Chromatin
47. The negative enhancers are called Silencers
48. The ability of embryonic cells to change their fates is called..... Regulation
49. Pax6 transcription factor found in mammalian eye, nervous system and pancreas.
50. E- Cadherin is expressed on all early mammalian embryonic cells.
51. Annelids, molluscs and flatworms belong to..... Lophotrochozoa
52. Potency of cell decrease if cell is committed.
53. Fertilization in Rana Pipiens usually lasts for about 3 (three) years.
54. An uncommitted cell can be described as Naive
55. Lying immediately beneath the plasma membrane of the egg is a thin Shell
56. Acroomal vesicles are considered as modified secretory vesicles
57. Most marine organisms release their gametes into the environment
58. The fast block method is used to prevent Polyspermy
59. Autonomous specification was first demonstrated in 1887
60. In mouse ZP3 is attached to zona pellucid
61. TAFs used in transcription factor activity
62. in slow block method which is cortical granule protein hyalin and peroxidase
63. Egg shell formed weight of egg. 9 – 12%
64. potency is intrinsic property
65. cytoplasm that contains many nuclei called syncytium
66. Ectoderm forms..... epidermis, nervous system, and pigment cells.
67. according to dutch sperm are ... parasitic animals (spermatozoa)
68. Acrosome & nucleus constitute the head of the sperm.
69. cortical granules are membrane bound structures
70. Acrosomal reaction has 2 (two) component
71. Acrosome-reacted sperm transfer their binding from ZP3 to the adjacent ZP2 molecules
72. Ca & bicarbonate ions maybe critical in activating cAMP

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73. Peroxidase enzyme hardens fertilization envelope
74. The Fast block to polyspermy is achieved by changing the electric potential
75. P-cadherin stands for _____Placental Cadherin
76. Most transcription factor can bind to specific DNA sequences
77. MITF protein necessary for production of pigments cell and their pigments.
78. Outside of plasma membrane is the vitelline envelope.
79. Mouse __fertilin___is localized to posterior plasma membrane of the sperm.
80. Finally the fourth cortical granules protein ___hyaline___ forms a coating around egg.
81. The cleavage stages are called.____Blastomere
82. Enhancers are....._DNA Sequences
83. The mitotic spindle and contractile ring are perpendicular to each other
84. J.Whittaker stained blastomeres for the presence of the enzyme acetyl cholinesterase
85. Protocadherins are involved in separating notochord from mesoderm in __Xenopus gastrulation
86. Pax6 factor needed for mammalian eye
87. Which stands comparative embryology is ____First_
88. Cell type specification precedes any large-scale embryonic cell migration
89. Series of extremely rapid divisions in zygote is ___Cleavage
90. In protosome, mouth is formed first than Deuterostomes: forms.
91. Northern blots, in situ hybridization, & the PCR can show which cells are transcribing particular genes.
92. Coding regions of DNA are called ___Exon
93. Na and __K__ Are important in fast block polyspermy.
94. The force for sperm propulsion is provided by __dynein
95. Spallenazani demonstrated that filtered __toad__ semen devoid of sperm would not fertilize egg.
96. Mesoderm generates the____KIDNEY.
97. Sperms are attracted towards egg of their species by CHEMOTAXIS.
98. Mathematical modeling seeks to desire developmental phenomenon in terms of EQUATIONS.
99. A region that is not translated in proteins is called UNTRANSLATED region.
100. Invariable cleavage produce the ___SAME___ lineages in embryo
101. Germ layers are formed as a result of ___GASTRULATION
102. RNA polymerase binds with PROMOTOR in start of transcription
103. The region of globular actin molecule lies between nuclear and acrosomal vesicle in ____SEA URCHIN
104. Which technique is used in situ hybridization ____Northern blots
105. Conditional specification involves interaction with RELATIVE POSITIONS cells
106. Particular proteins or carbohydrate on the sperm surface __LOST__ during capacitation
107. ___LOSS____ of hydrogen ion causes pH to rise.
108. Peroxidase enzyme hardens the fertilization envelope by crosslinking TYROSINE residues
109. FERTILIZATION transmits genes from parents to offspring.
110. Regulation of gene expression accomplished by SELECTIVE MESSENGER RNA TRANSLATION
111. Development contact genotype and PHENOTYPE
112. Nucleosome is composed of OCTAMER of histone protein.
113. Knowledge of gene activity in humans can be obtained by candidate GENE MAPPING & POSITIONAL CLONING
114. Anatomical abnormalities are caused by MUTANT GENES
115. All multicellular organisms arise by a slow process of progressive change called DEVELOPMNET

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116. Animals development is the division of embryo into three layers Ectoderm, Mesoderm and ENDODERM
117. The ions which are responsible for increasing granule reactions Ca (Calcium)
118. The zona pellucida is a fglycoprotein matrix, which is synthesized and secreted by the growing oocyte
119. In several other species, certain regions of the membrane are specialized for sperm recognition & fusion
120. An intrinsic program has been activated within the cell that causes it to follow a particular pathway of development
121. Activation of egg metabolism to start development
122. Insects such as Drosophila use all three modes of specification to commit their cells to particular fates
123. Nuclear control of cell morphogenesis and the interaction of nucleus and cytoplasm are well studied in Acetabularia.
124. Upon fertilization, the intracellular calcium ion concentration of the egg increases greatly
125. Bacteria are able to transmit genes from one individual to another by means of sex pili
126. +ve and – ve mating take place *Chlamydomonas*

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SUBJECTIVE

2marks:

1. **Define Blastocoel?**

A fluid-filled cavity, the blastocoel, forms in the animal hemisphere. This cavity will be important for allowing cell movements to occur during gastrulation.

2. **Define periosteum?**

The entire region of calcified spicules becomes surrounded by compact mesenchymal cells that form the **Periosteum** (a membrane that surrounds the bone).

3. **Define Germ ring?**

After the blastoderm cells have covered about half the zebra fish yolk cell, a thickening occurs throughout the margin of the epibolizing blastoderm. This thickening is called the **germ ring**, and it is composed of a superficial layer, the **epiblast**, and an inner layer, the **hypoblast**.

4. **What is primitive knot or hensen node?**

At the anterior end of the primitive streak is a regional thickening of cells called the **primitive knot** or **Hensen's node.**

5. **Write difference between Bicoid and Nanos?**

The anterior most portion of the egg contains an mRNA that encodes a protein called Bicoid. The posterior most portion of the egg contains an mRNA that encodes a protein called Nanos.

6. **Difference between axoneme & dyenin?**

Answer: The major motor portion of the flagellum is called the **axoneme.**

The force for sperm propulsion is provided by **dynein**, a protein that is attached to the microtubules.

7. **Define Silencers?**

Enhancers can also be used to inhibit transcription. In some cases, the same transcription factors that activate the transcription of one gene can be used to repress the transcription of other genes. These "negative enhancers " are also called silencers.

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8. **What is embryonic shield?**

The cells of both the epiblast and hypoblast intercalate on the future dorsal side of the embryo to form a localized thickening, the embryonic shield.

9. **What is naïve?**

An uncommitted cell can be described as naive, meaning that it has received no instructions directing it along a particular developmental pathway.

10. **Egg jelly formation?**

Many types of eggs have glycoprotein meshwork called egg jelly outside the vitelline envelope which is used either to attract or to activate sperm. The egg, then, is a cell specialized for receiving sperm and initiating development.

11. **Differentiate between Blastomere and cleavage?**

Cleavage is a series of extremely rapid mitotic divisions wherein the enormous volume of zygote cytoplasm is divided into numerous smaller cells. These cells are called **blastomeres.**

12. **What is marginal belt?**

Between the area pellucida and the area opaca is a thin layer of cells called the **marginal zone** (or **marginal belt**).

13. **Define Neural Crest?**

A group of ectodermal cells that gives rise to the spinal ganglia and various structures of the autonomic nervous system.The neural crest cells originate at the dorsal most region of the neural tube.The neural crest can be divided into four main functional (but overlapping) domains.

14. **Define Regulative development?**

conditional specification gives rise to a pattern of embryogenesis called **regulative development**.

15. **Neural plate in tunicats?**

In **primary neurulation**, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.

16. **Role.of TGF nd FGF7 in human body development**

Paracrine factors such as **TGF- and FGF7** are important in normal skin development.

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17. **Define primary and secondary neurulation?**

In **primary neurulation**, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.

In **secondary neurulation**, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube. **Neurulation in fishes is exclusively secondary.**

18. **Define Prechordal plate?**

Cells migrating through Hensen's node become chordamesoderm (notochord) cells. These extend up to the presumptive midbrain, where they meet the prechordal plate. The **prechordal plate** induces the formation of the forebrain.

3MARKS

19. **Write three distinct lineage of Osteogenesis? Or What are lineages to generate skeleton?**

There are three distinct lineages that generate the skeleton.

- The somites generate the axial skeleton.
- The lateral plate mesoderm generates the limb skeleton.
- The cranial neural crest gives rise to the branchial arch and craniofacial bones and cartilage.

20. **What are syncytiotrophoblast and its function?**

The multinucleated type of cell forms the **syncytiotrophoblast**.

The extra-embryonic mesoderm joins the trophoblastic extensions and gives rise to the blood vessels that carry nutrients from the mother to the embryo. The narrow connecting stalk of extra-embryonic mesoderm that links the embryo to the trophoblast eventually forms the vessels of the **umbilical cord**.

21. **Write the function of Metencephalon?**

The Myelencephalon eventually becomes the **Medulla oblongata**, whose neurons generate the nerves that regulate respiratory, gastrointestinal, and cardiovascular movements. The **Metencephalon** gives rise to the cerebellum, the part of the brain responsible for coordinating movements, posture, and balance.

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22. Pattern of embryonic cleavage?

Patterns of embryonic cleavage

Indeed, different organisms undergo cleavage in distinctly different ways. The pattern of embryonic cleavage particular to a species is determined by two major parameters:

1. The amount and distribution of yolk protein within the cytoplasm.
2. The factors in the egg cytoplasm that influence the angle of the mitotic spindle and the timing of its formation.
3. The yolk-rich pole is referred to as the **vegetal pole**; the yolk concentration in the **animal pole** is relatively low. The zygote nucleus is frequently displaced toward the animal pole. In general, yolk inhibits cleavage.

23. Write the role of calcium in the fusion of egg and sperm?

Upon fertilization, the intracellular calcium ion concentration of the egg increases greatly. In this high-calcium environment, the cortical granule membranes fuse with the egg plasma membrane, releasing their contents. Once the fusion of the cortical granules begins near the point of sperm entry, a wave of cortical granule exocytosis propagates around the cortex to the opposite side of the egg. The release of calcium from intracellular storage can be monitored visually using calcium-activated luminescent dyes such as aequorin (isolated from luminescent jellyfish) or fluorescent dyes such as fura-2. These dyes emit light when they bind free calcium ions.

Several experiments have demonstrated that calcium ions are directly responsible for propagating the cortical granule reaction, and that these calcium ions are stored within the egg itself.

24. Differentiate between Autonomous and syncytial specification?

Autonomous specification:

Specification by differential acquisition of certain cytoplasmic molecules present in the egg.

- Characteristic of most invertebrates.
- Cell type specification precedes any large-scale embryonic cell migration.
- Autonomous specification gives rise to a pattern of development referred to as **mosaic development**, in which the embryo appears to be constructed like a tile mosaic of independent self-differentiating parts. Invariant cleavages produce the same lineages in each embryo of the species. Blastomere fates are generally invariant.

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Syncytial specification:

- Characteristic of most insect classes.
- Specification of body regions by interactions between cytoplasmic regions prior to cellularization of the blastoderm.
- Variable cleavage produces no rigid cell fates for particular nuclei.
- After cellularization, conditional specification is most often seen.

25. Difference between evolutionary embryology and tetralogy?

Evolutionary embryology:

The second strand, based on the first, is evolutionary embryology, the study of how changes in development may cause evolutionary changes and of how an organism's ancestry may constrain the types of changes that are possible.

Teratology:

The third anatomical approach to developmental biology is **teratology**, the study of birth defects. These anatomical abnormalities may be caused by mutant genes or by substances in the environment that interfere with development. The study of abnormalities is often used to discover how normal development occurs.

26. Difference b/w identical and fraternal twins?

Identical twins occur in roughly **0.25%** of human births. About **33%** of identical twins have two complete and separate chorions, indicating that separation occurred before the formation of the trophoblast tissue at **day 5.**

Fraternal twins are the result of two separate fertilization events, whereas identical twins are formed from a single embryo whose cells somehow dissociated from one another.

27. Differnce b/w gray matter and white matter?

Glial cells cover many of the axons in the marginal zone in myelin sheaths, giving them a whitish appearance. Hence, the mantle zone, containing the neuronal cell bodies, is often referred to as the **Gray matter**;

The axonal, marginal layer is often called the **White matter**. In the spinal cord and medulla, this basic three-zone pattern of ependymal, mantle, and marginal layers is retained throughout development. The gray matter (mantle) gradually becomes a butterfly-shaped structure surrounded by white matter; and both become encased in connective tissue.

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28. Define Medial hinge point?

In birds and mammals, the cells at the midline of the neural plate are called the **medial hinge point** (**MHP**) **cells**. They are derived from the portion of the neural plate just anterior to Hensen's node and from the anterior midline of Hensen's node.

29. Diff b/w late and early response of egg to sperm?

Early responses: Contact b/w sea urchin sperm & egg activates the 2 major blocks to polyspermy: the fast block, initiated by sodium influx into the cell, & the slow block, initiated by the intracellular release of Ca+ ions. The activation of all eggs appears to depend on an increase in the conc. of free Ca+ ions within the egg. Such an increase can occur in 2 ways: Ca+ ions can enter the egg from outside, or Ca+ ions can be released from the endoplasmic reticulum within the egg.
Late responses: Shortly after the Ca+ ion levels rise in a sea urchin egg, its intracellular pH also increases. The rise in intracellular pH begins with a 2nd influx of Na+ ions, which causes a 1:1 exchange b/w sodium ions from the seawater & hydrogen ions from the egg. This loss of hydrogen ions causes the pH to rise. It is thought that the pH increase and the calcium ion elevation act together to stimulate new protein synthesis and DNA synthesis.

30. Write the general concept of fertilization?

Fertilization is the process whereby two sex cells (gametes) fuse together to create a new individual with genetic potentials derived from both parents.

Fertilization accomplishes two separate processes:

- Sex (the combining of genes derived from the two parents) and
- Reproduction (the creation of new organisms)
- Thus, the first function of fertilization is to transmit genes from parent to offspring, and the second is to initiate in the egg cytoplasm those reactions that permit development to proceed.

31. What is Marble bone disease?

Osteopetrosis or Marble bone disease

Osteopetrosis is a bone disease that makes bones abnormally dense and prone to breakage (fracture).

This disorder is characterized by increased bone density and abnormal bone growth.

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32. Estrogen and progesterone function?

Estrogen is a hormone that can instruct the liver to make and secrete the yolk proteins, which are then transported through the blood into the enlarging eggs in the ovary. The yolk is transported into the bottom portion of the egg.

progesterone, signals the egg to resume its meiotic division. This is necessary because the egg had been "frozen" in the metaphase of its first meiosis. When it has completed this first meiotic division, the egg is released from the ovary and can be fertilized.

33. What are resact and its function?

Answer: **Resact** is specific for ***A. punctulata*** and does not attract sperm of other species. *A. punctulata* sperm have receptors in their plasma membranes that bind resact and can swim up a concentration gradient of this compound until they reach the egg. Resact also acts as a **sperm-activating peptide**.

5MARKS:

34. Write the functional domains of neural crest?

- A group of ectodermal cells that gives rise to the spinal ganglia and various structures of the autonomic nervous system.
- The neural crest cells originate at the dorsal most region of the neural tube.
- The neural crest can be divided into four main functional (but overlapping) domains.
- The cranial (cephalic) neural crest, whose cells migrate dorso-laterally to produce the craniofacial mesenchyme that differentiates into the cartilage, bone, cranial neurons, glia, and connective tissues of the face.

35. life cycle of chlamydomonas?

the life cycle of *Chlamydomonas.* Which is usually haploid, having just one copy of each chromosome. The individuals of each species, however, are divided into two mating types: plus and minus. When a plus and a minus meet, they join their cytoplasm, and their nuclei fuse to form a diploid zygote. The flagella of two individuals twist around each other, enabling specific regions of the cell membranes to come together. These specialized regions contain mating type-specific components that enable the cytoplasms to fuse.

36. Describe primary neurulation?

Primary neurulation

- During primary neurulation, the original ectoderm is divided into three sets of cells.
- The internally positioned neural tube, which will form the brain and spinal cord.

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- The externally positioned epidermis of the skin.
- The neural crest cells. The neural crest cells form in the region that connects the neural tube and epidermis, but then migrate elsewhere; they will generate the peripheral neurons and glia, the pigment cells of the skin, and several other cell types.

37. Transcription domain?

Transcription factors have three major domains.

1. **DNA-binding domain**
2. **Protein-protein interaction domain**
3. ***Trans*-activating domain**

- **DNA-binding domain:**It recognizes a particular DNA sequence.
- **Protein-protein interaction domain:** It allows the transcription factor's activity to be modulated by TAFs or other transcription factors.
- ***Trans*-activating domain:** It activates or suppresses the transcription of the gene whose promoter or enhancer it has bound. Usually, this *trans*-activating domain enables the transcription factor to interact with proteins involved in binding RNA polymerase.Examples,***MITF:(***Microphthalmia-Associated Transcription Factor. The microphthalmia (MITF) protein is necessary for the production of pigment cells and their pigments.
- ***Pax6:*** Paired box gene 6. The Pax6 transcription factor, which is needed for mammalian eye, nervous system, and pancreas development, contains two potential DNA-binding domains.
- **Pdx1:** Pancreatic And Duodenal Homeobox 1 gene. It is specific for the pancreatic region of the endoderm.

38. 2 ways for the formation of neural tube?

Formation of the Neural Tube

There are two major ways of forming a neural tube

1) Primary neurulation 2) **Secondary neurulation**

- In **primary neurulation**, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.

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- In **secondary neurulation**, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube. **Neurulation in fishes is exclusively secondary.**
- In **amphibians**, such as ***Xenopus,*** most of the tadpole neural tube is made by **primary neurulation**, but the tail neural tube is derived from **secondary neurulation**.

39. Enlist 4 factors that cause movement in gene regulation?

The regulation of gene expression can be accomplished at several levels:

- **Differential gene transcription:** It regulates that which of the nuclear genes are transcribed into RNA.
- **Selective nuclear RNA processing:** It regulating which of the transcribed RNAs (or which parts of such a nuclear RNA) enter into the cytoplasm to become messenger RNAs.
- **Selective messenger RNA translation:** It regulates that which of the mRNAs in the cytoplasm become translated into proteins.
- **Differential protein modification:** It regulates that which proteins are allowed to remain or function in the cell.

40. Write modes of commitment?

Three basic modes of commitment are

1: Autonomous specification:

Specification by differential acquisition of certain cytoplasmic molecules present in the egg.

- Characteristic of most invertebrates.
- Cell type specification precedes any large-scale embryonic cell migration.
- Autonomous specification gives rise to a pattern of development referred to as **mosaic development**, in which the embryo appears to be constructed like a tile mosaic of independent self-differentiating parts. Invariant cleavages produce the same lineages in each embryo of the species. Blastomere fates are generally invariant.

Produces "mosaic" ("determinative") development: cells cannot change fate if a blastomere is lost.

2: Conditional specification:

- Characteristic of all vertebrates and few invertebrates.

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- Specification by interactions between cells. Relative positions are important.
- Variable cleavages produce no invariant fate assignments to cells.
- Massive cell rearrangements and migrations precede or accompany specification.

Capacity for "regulative" development: allows cells to acquire different functions.

3: Syncytial specification:

- Characteristic of most insect classes.
- Specification of body regions by interactions between cytoplasmic regions prior to cellularization of the blastoderm.
- Variable cleavage produces no rigid cell fates for particular nuclei.
- After cellularization, conditional specification is most often seen.

41. Define : Area opaca , Area pellicuda , sub-germinal cavity, marginal zone?

- Between the blastoderm and the yolk is a space called the **sub-germinal cavity**.
- The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the **area opaca**.
- The deep cells in the center of the blastoderm are shed and die, leaving behind a onecell- thick **area pellucida**.
- Between the area pellucida and the area opaca is a thin layer of cells called the **marginal zone** (or **marginal belt**).

42. Hammering experiment for role of nucleus in morphology?

Nuclear control of cell morphogenesis and the interaction of nucleus and cytoplasm are well studied in *Acetabularia.* This enormous single cell (2-4 cm long) consists of three parts:

- a cap,
- a stalk,
- a rhizoid.

The single nucleus of the cell resides within the rhizoid. The size of *Acetabularia* and the location of its nucleus allow investigators to remove the nucleus from one cell and replace it with a nucleus from another cell.

J.Hammerling exchanged nuclei between two morphologically distinct species, *A. mediterranea* and *A. crenulata* **J. Hammerling** found that when he transferred the nucleus from one species into the stalk of another species, the newly

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formed cap eventually assumed the form associated with the *donor* nucleus. Thus, the nucleus was seen to control *Acetabularia* development.

The formation of a cap is a complex morphogenic event involving the synthesis of numerous proteins, which must be accumulated in a certain portion of the cell and then assembled into complex, species-specific structures.

10 Marks:

43. Endochondral?

- Endochondral ossification involves the formation of cartilage tissue from aggregated mesenchymal cells, and the subsequent replacement of cartilage tissue by bone.
- The process of Endochondral ossification can be divided into five stages.
- First, the mesenchymal cells are committed to become cartilage cells. This commitment is caused by paracrine factors that induce the nearby mesodermal cells to express two transcription factors, Pax1 and **Scleraxis**.
- Pax1 and Scleraxis are the transcription factors are thought to activate cartilage-specific genes.
- Thus, Scleraxis is expressed in the mesenchyme from the sclerotome, in the facial mesenchyme that forms cartilaginous precursors to bone, and in the limb mesenchyme
- During the second phase of endochondral ossification, the committed mesenchyme cells condense into compact nodules and differentiate into **chondrocytes**, the cartilage cells. N-cadherin appears to be important in the initiation of these condensations.
- During the second phase of endochondral ossification, the committed mesenchyme cells condense into compact nodules and differentiate into **chondrocytes**, the cartilage cells
- In humans, the *SOX9* gene, which encodes a DNA-binding protein, is expressed in the precartilaginous condensations. Mutations of the *SOX9* gene cause camptomelic dysplasia, a rare disorder of skeletal development that results in deformities of most of the bones of the body.
- During the third phase of endochondral ossification, the chondrocytes proliferate rapidly to form the model for the bone. As they divide, the chondrocytes secrete a cartilage-specific extracellular matrix In the fourth phase, the chondrocytes stop dividing and increase their volume dramatically, becoming hypertrophic chondrocytes. These large chondrocytes alter the matrix they produce (by adding collagen X and more fibronectin) to enable it to become mineralized by calcium carbonate.
- The fifth phase involves the invasion of the cartilage model by blood vessels.
- **The hypertrophic chondrocytes die by apoptosis.** This space will become **bone marrow**. As the cartilage cells die, a group of cells that have surrounded the cartilage model differentiate into osteoblasts.
- The osteoblasts begin forming bone matrix on the partially degraded cartilage. Eventually, all the cartilage is replaced by bone. Thus, the cartilage tissue serves as a model for the bone that follows

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44. Explain Osteogenesis in detail?

Osteogenesis:

- Some of the most obvious structures derived from the paraxial mesoderm are bones.

There are three distinct lineages that generate the skeleton.

- The somites generate the axial skeleton.
- The lateral plate mesoderm generates the limb skeleton.
- The cranial neural crest gives rise to the branchial arch and craniofacial bones and cartilage.
- There are two major modes of bone formation, or **osteogenesis**, and both involve the transformation of a preexisting mesenchymal tissue into bone tissue.
- The direct conversion of mesenchymal tissue into bone is called **intramembranous ossification**. This process occurs primarily in the bones of the skull.
- In other cases, the mesenchymal cells differentiate into cartilage, and this cartilage is later replaced by bone. The process by which a cartilage intermediate is formed and replaced by bone cells is called endochondral ossification.

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ZOO501-QUIZ 1, FALL 21

ATTEMPTED BY

Muhammad Kashif

03064963933

ZOO501 - Developmental Biology (Quiz # 1)

Question # 1 of 10 (Start time: 12:28:13 AM, 19 December 2021)

The _____ factors that initiate the acrosomal reaction in sea urchins are often highly

Select the correct option

- (●) Egg jelly
- ( ) Sperm
- ( ) Ovulation
- ( ) Zona

+empted by Muhammad Kashif

12:29 445 B/s 100

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ZOO501 - Developmental Biology (Quiz # 1)

Question # 2 of 10 (Start time: 12:29:00 AM, 19 December 2021)

_____ is the process whereby two sex cells (gametes) fuse together to create a new i
both parents.

Select the correct option

- Genetic Recombination
- Fertilization
- Crossing Over
- Cloning

Attempted by Muhammad Kashif

12:30 198 B/s 100

ZOO501 - Developmental Biology (Quiz # 1) Qu

Question # 3 of 10 (Start time: 12:29:43 AM, 19 December 2021)

The exocytosis of the acrosomal vesicle is caused by the calcium-mediated fusion of the acrosomal m plasma membrane?

Select the correct option

- (●) True
- ( ) False

Click to

Attempted by Muhammad Kashif

12:31 119 B/s 100

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ZOO501 - Developmental Biology (Quiz # 1) Quiz Start Tir

Question # 4 of 10 (Start time: 12:30:50 AM, 19 December 2021)

The first evidence suggesting the importance of sperm in reproduction came from a series of experiments perforr

Select the correct option

- J. L. Prevost
- G. J. Mendel
- Lazzaro Spallanzani
- J. B. Dumas

Click to Save Answ

Attempted by Muhammad Kashif

12:32 424 B/s 100

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ZOO501 - Developmental Biology (Quiz # 1)

Question # 5 of 10 (Start time: 12:31:43 AM, 19 December 2021)

Major Events in the process of Fertilization are _____.

Select the correct option

- 6
- 2
- 4
- 8

Attempted by Muhammad Kashif

12:32 408 B/s 100

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ZOO501 - Developmental Biology (Quiz # 1)

Question # 6 of 10 (Start time: 12:32:15 AM, 19 December 2021)

Which transmit genes from parents to offspring?

Select the correct option

- Development
- Fermentation
- Fertilization
- Cleavage

Attempted by Muhammad Kashif

12:33 vulms.vu.edu.pk/Quiz/(

Question # 7 of 10 (Start time: 12:33:04 AM, 19 December 2021)

The first evidence suggesting the importance of sperm in reproduction came from a series of experim
the late _____s. Spallanzani demonstrated that filtered toad semen devoid of sperm would not fertili

Select the correct option

- 1900
- 1800
- 1700
- 1600

Click to

Attempted by Muhammad Kashif

12:34 vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1)

Question # 8 of 10 (Start time: 12:33:39 AM, 19 December 2021)

The exocytosis of the acrosomal vesicle is caused by the _____-mediated fusion of the acrosoma plasma membrane?

Select the correct option

- Calcium
- Sodium
- Magnesium
- Potassium

Click

Attempted by Muhammad Kashif

12:34 17 B/s 100

vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1) Quiz Start Time

Question # 9 of 10 (Start time: 12:34:16 AM, 19 December 2021)

A.V Leeuwenhoek, the Dutch microscopist who co-discovered sperm in _____

Select the correct option

- 1878
- 1778
- 1978
- 1678

Click to Save Answe

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12:35 538 B/s 100

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ZOO501 - Developmental Biology (Quiz # 1) Quiz

Question # 10 of 10 (Start time: 12:34:54 AM, 19 December 2021)

The shell is the egg's outer covering, accounting for about 9 to 12% of its tota weight depending on egg siz defense against bacterial contamination. This bumpy and grainy outer covering has approximately ____ entirely of calcium carbonate?

Select the correct option

- 170
- 17000
- 1700
- 17

Click to Sa

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12:37 vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1) Qui

Question # 1 of 10 (Start time: 12:37:00 AM, 19 December 2021)

Sperm consists of _____.

Select the correct option

- A sac of enzymes
- All of these
- A haploid nucleus
- A propulsion system

Click to

Attempted by Muhammad Kashif

12:38 103 B/s 100

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ZOO501 - Developmental Biology (Quiz # 1) Quiz St

Question # 2 of 10 (Start time: 12:37:53 AM, 19 December 2021)

Which transmit genes from parents to offspring?

Select the correct option

- (●) Fertilization
- ( ) Fermentation
- ( ) Development
- ( ) Cleavage

Click to Save

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12:38 320 B/s 100

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ZOO501 - Developmental Biology (Quiz # 1) Quiz

Question # 3 of 10 (Start time: 12:38:10 AM, 19 December 2021)

The _____ diffuses readily in seawater and has a profound effect at very low concentrations when sperm.

Select the correct option

- Trisect
- Resact
- Presact
- None

Click to Sc

Attempted by Muhammad Kashif

12:39 382 B/s 100

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ZOO501 - Developmental Biology (Quiz # 1) Quiz Star

Question # 4 of 10 (Start time: 12:38:41 AM, 19 December 2021)

The major motor portion of the flagellum is called _____.

Select the correct option

- Seminal Vesicla
- axoneme
- Acrosome
- acrosomal vesicle

Click to Save A

Attempted by Muhammad Kashif

12:39 502 B/s 100

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ZOO501 - Developmental Biology (Quiz # 1) Quiz S

Question # 5 of 10 (Start time: 12:39:15 AM, 19 December 2021)

The exocytosis of the acrosomal vesicle is caused by the calcium-mediated fusion of the acrosomal mem plasma membrane?

Select the correct option

- (●) True
- ( ) False

Click to Sav

Attempted by Muhammad Kashif

12:40 vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1) Quiz Start Ti

Question # 7 of 10 (Start time: 12:40:09 AM, 19 December 2021)

The first evidence suggesting the importance of sperm in reproduction came from a series of experiments perfo the late _____s. Spallanzani demonstrated that filtered toad semen devoid of sperm would not fertilize eggs?

Select the correct option

- 1900
- 1800
- 1600
- 1700

Click to Save Ans

Attempted by Muhammad Kashif

12:41 485 B/s 100

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ZOO501 - Developmental Biology (Quiz # 1) Quiz S

Question # 8 of 10 (Start time: 12:40:55 AM, 19 December 2021)

In egg, outside the plasma membrane is the vitelline envelope, which forms a fibrous mat around the egg _____ different glycoproteins and is often involved in sperm-egg recognition.

Select the correct option

- 5
- 7
- 8
- 6

Click to Sa

Attempted by Muhammad Kashif

12:42 99 B/s 100

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ZOO501 - Developmental Biology (Quiz # 1) Quiz Star

Question # 9 of 10 (Start time: 12:42:01 AM, 19 December 2021)

Miller found that when sperm were added to oocytes that had not yet completed their _____ meiotic di
sperm to eggs?

Select the correct option

- 3
- 4
- 2
- 1

Click to Save A

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12:43 10 K/s 100

vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1)

Question # 10 of 10 (Start time: 12:42:48 AM, 19 December 2021)

Protein synthesis is accomplished by ribosomes and _____

Select the correct option

- DNA
- rRNA
- tRNA
- None of above

Attempted by Muhammad Kashif

12:46 59 B/s 100

ZOO501 - Developmental Biology (Quiz # 1) Quiz Start T

Question # 1 of 10 (Start time: 12:45:48 AM, 19 December 2021)

Miller found that when sperm were added to oocytes that had not yet completed their _____ meiotic divis sperm to eggs?

Select the correct option

- 3
- 1
- 2
- 4

Click to Save An

Attempted by Muhammad Kashif

12:46

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ZOO501 - Developmental Biology (Quiz # 1) Quiz Start Time

Question # 2 of 10 (Start time: 12:46:09 AM, 19 December 2021)

The major motor portion of the flagellum is called _____.

Select the correct option

- axoneme
- Seminal Vesicla
- acrosomal vesicle
- Acrosome

Click to Save Answe

Attempted by Muhammad Kashif

12:47 94 B/s 100

vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1) Quiz Star

Question # 3 of 10 (Start time: 12:46:55 AM, 19 December 2021)

Which transmit genes from parents to offspring?

Select the correct option

- Development
- Fertilization
- Fermentation
- Cleavage

Click to Save

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12:47 vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1) Quiz Start

Question # 4 of 10 (Start time: 12:47:34 AM, 19 December 2021)

Protein synthesis is accomplished by ribosomes and _____

Select the correct option

- (●) tRNA
- ( ) None of above
- ( ) DNA
- ( ) rRNA

Click to Save A

Attempted by Muhammad Kashif

12:48 232 B/s 100

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ZOO501 - Developmental Biology (Quiz # 1)

Question # 5 of 10 (Start time: 12:47:50 AM, 19 December 2021)

MITF is required for _____.

Select the correct option

- for the production of germ cells .
- for the production of Non-pigmented cells.
- the production of Nucleated cells.
- the production of pigment cells and their pigments.

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12:49 vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1) Quiz Star

Question # 6 of 10 (Start time: 12:48:56 AM, 19 December 2021)

The first evidence suggesting the importance of sperm in reproduction came from a series of experiments pe
the late _____s. Spallanzani demonstrated that filtered toad semen devoid of sperm would not fertilize eggs

Select the correct option

- 1900
- 1700
- 1800
- 1600

Click to Save

Attempted by Muhammad Kashif

12:51 452 B/s 100

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ZOO501 - Developmental Biology (Quiz # 1) Quiz Start Time: 12

Question # 8 of 10 (Start time: 12:51:04 AM, 19 December 2021)

Which Transcription factor is required for mammalian eye, nervous system, and pancreas development?

Select the correct option

- MITF
- Pax6
- None
- Pdx1

Click to Save Answer &

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12:52 400 B/s 100

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ZOO501 - Developmental Biology (Quiz # 1)

Question # 9 of 10 (Start time: 12:51:51 AM, 19 December 2021)

A.V Leeuwenhoek, the Dutch microscopist who co-discovered sperm in _____

Select the correct option

- (●) 1678
- ( ) 1778
- ( ) 1978
- ( ) 1878

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12:52 815 B/s 100

vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1) Quiz Sta

Question # 10 of 10 (Start time: 12:52:26 AM, 19 December 2021)

The region of globular actin molecules lies between the nucleus and the acrosomal vesicle in _____

Select the correct option

- Rat
- Chick
- Frog
- sea urchins

Click to Save

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12:54 vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1) Quiz S

Question # 1 of 10 (Start time: 12:54:36 AM, 19 December 2021)

The exocytosis of the acrosomal vesicle is caused by the calcium-mediated fusion of the acrosomal mem plasma membrane?

Select the correct option

- (●) True
- ( ) False

Click to Sa

Attempted by Muhammad Kashif

12:55 503 B/s 100

vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1)

Question # 2 of 10 (Start time: 12:54:48 AM, 19 December 2021)

The first evidence suggesting the importance of sperm in reproduction came from a series of experim

Select the correct option

- J. B. Dumas
- G. J. Mendel
- Lazzaro Spallanzani
- J. L. Prevost

Click t

Attempted by Muhammad Kashif

12:56 vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1) Qui

Question # 3 of 10 (Start time: 12:55:23 AM, 19 December 2021)

The second part of the acrosomal reaction involves the extension of the acrosomal process. This protru of globular actin molecules into actin filaments?

Select the correct option

- (•) True
- ( ) False

Click to

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12:57 vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1) Quiz Start Time: 12:

Question # 4 of 10 (Start time: 12:56:26 AM, 19 December 2021)

Which Transcription factor is required for mammalian eye, nervous system, and pancreas development?

Select the correct option

- Pdx1
- None
- MITF
- Pax6

Click to Save Answer &

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12:58 386 B/s 99

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ZOO501 - Developmental Biology (Quiz # 1)

Question # 5 of 10 (Start time: 12:57:39 AM, 19 December 2021)

_____ is the process whereby two sex cells (gametes) fuse together to create a new indivic both parents.

Select the correct option

- Genetic Recombination
- Cloning
- Crossing Over
- Fertilization

Click

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12:58 ⚠ ⚙ 📰 • 4G 1 K/s 99

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ZOO501 - Developmental Biology (Quiz # 1) Quiz

Question # 6 of 10 (Start time: 12:58:13 AM, 19 December 2021)

MITF is required for _____.

Select the correct option

- the production of Nucleated cells.
- for the production of germ cells .
- the production of pigment cells and their pigments.
- for the production of Non-pigmented cells.

Click to S

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12:59 vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1) Quiz St

Question # 7 of 10 (Start time: 12:58:45 AM, 19 December 2021)

The mammalian egg is also surrounded by a layer of cells called _____

Select the correct option

- Coat
- Cumulus
- Cell wall
- All of above

Click to Save

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1:00 446 B/s 99

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ZOO501 - Developmental Biology (Quiz # 1) Quiz Start Time: 1

Question # 8 of 10 (Start time: 12:59:20 AM, 19 December 2021)

The _____ factors that initiate the acrosomal reaction in sea urchins are often highly specific to each species?

Select the correct option

- Ovulation
- Sperm
- Zona
- Egg jelly

Click to Save Answer

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1:00 68 K/s 99

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ZOO501 - Developmental Biology (Quiz # 1) Quiz Sta

Question # 9 of 10 (Start time: 01:00:15 AM, 19 December 2021)

Contact with egg jelly causes the _____ of the sperm's acrosomal vesicle and the release of proteolytic en through the jelly coat to the egg surface?

Select the correct option

- (●) Exocytosis
- ( ) Palleocytosis
- ( ) Endocytosis
- ( ) None

Click to Save

Attempted by Muhammad Kashif

1:01 457 B/s 99

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ZOO501 - Developmental Biology (Quiz # 1) Quiz

Question # 10 of 10 (Start time: 01:00:59 AM, 19 December 2021)

Only one sperm can ultimately fertilize the egg.

Select the correct option

- (●) True
- ( ) False

Click to S

Attempted by Muhammad Kashif

1:04 888 B/s 98

vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1) Quiz Sta

Question # 1 of 10 (Start time: 01:03:24 AM, 19 December 2021)

The mechanisms of chemotaxis differ among species. One chemotactic molecule, a 14-amino acid peptide isolated from the egg jelly of the sea urchin?

Select the correct option

- Presect
- Trisect
- Resect
- None

Click to Save

Attempted by Muhammad Kashif

1:05 391 B/s 98

vulms.vu.edu.pk/Quiz/(

Question # 2 of 10 (Start time: 01:04:18 AM, 19 December 2021)

The innermost layer of cumulus cells, immediately adjacent to the zona pellucida, is call

Select the correct option

- (●) Corona radiata
- ( ) Zona pellucida
- ( ) Cumulus
- ( ) Vitelline envelope

Attempted by Muhammad Kashif

1:05 413 B/s 98

vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1)

Question # 3 of 10 (Start time: 01:05:27 AM, 19 December 2021)

The interaction of sperm and egg generally proceeds according to _____ basic

Select the correct option

- 3
- 5
- 4
- 2

Attempted by Muhammad Kashif

1:06 409 B/s 98

vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1)

Question # 4 of 10 (Start time: 01:06:04 AM, 19 December 2021)

The egg white is known as the Yolk?

Select the correct option

- True
- False

Click t

Attempted by Muhammad Kashif

1:07 389 B/s 98

vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1)

Question # 5 of 10 (Start time: 01:06:43 AM, 19 December 2021)

Gametes of _____ remains one of the best-studied examples of fertilization.

Select the correct option

- Hydra
- Star Fish
- Sea Cocumber
- Sea urchin

Attempted by Muhammad Kashif

1:07 430 B/s 98

vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1)

Question # 6 of 10 (Start time: 01:07:19 AM, 19 December 2021)

In many species, the meeting of sperm and egg is not a simple matter and faces _____problems?

Select the correct option

- 3
- 2
- 4
- 1

Click to

Attempted by Muhammad Kashif


1:07
vulms.vu.edu.pk/Quiz/(
ZOO501 - Developmental Biology (Quiz # 1)
Quiz Start Ti
Question # 7 of 10 (Start time: 01:07:39 AM, 19 December 2021)
Which transmit genes from parents to offspring?
Select the correct option
Cleavage
Development
Fertilization
Fermentation
Click to Save Ans
Attempted by Muhammad Kashif

1:08 vulms.vu.edu.pk/Quiz/(

Question # 8 of 10 (Start time: 01:07:53 AM, 19 December 2021)

Sperm consists of _____.

Select the correct option

- A propulsion system
- A sac of enzymes
- All of these
- A haploid nucleus

Click to Save

Attempted by Muhammad Kashif

1:08 ⚠ ⚙ 📰 • 4G 17 B/s 98

vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1)

Question # 9 of 10 (Start time: 01:08:07 AM, 19 December 2021)

In most marine invertebrates, the acrosomal reaction has ____ components?

Select the correct option

- 1
- 3
- 4
- 2

Attempted by Muhammad Kashif

1:09 469 B/s 97

vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1) Quiz Start Time: 01:03 AM

Question # 10 of 10 (Start time: 01:08:42 AM, 19 December 2021)

The second part of the acrosomal reaction involves the extension of the acrosomal process. This protrusion arises through th of globular actin molecules into actin filaments?

Select the correct option

- True
- False

Click to Save Answer & Move

Attempted by Muhammad Kashif

12:40 270 B/s 100

vulms.vu.edu.pk/Quiz/(

ZOO501 - Developmental Biology (Quiz # 1) Quiz

Question # 6 of 10 (Start time: 12:39:32 AM, 19 December 2021)

The interaction of sperm and egg generally proceeds according to _____ basic step?

Select the correct option

- 3
- 4
- 2
- 5

Click to S

Attempted by Muhammad Kashif

ZOO501
QUIZ 2, FALL 21

SOLVED BY
AYESHA
MERAJ

2:53 294 B/s 50
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 10 of 10 (Start time: 02:52:53 AM, 07 February 2022)

The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the ---

Select the correct option

- (●) Area opaca
- ( ) Area pellucid
- ( ) Marginal belt
- ( ) Marginal zone

Solved by Ayesha Meraj

2:52 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 9 of 10 (Start time: 02:52:21 AM, 07 February 2022)

The mammalian embryo obtains nutrients directly from its ----------

Select the correct option

- Stored yolk
- Animal pole
- Vegatal pole
- Mother

Solved by Ayesha Meraj

2:52 348 B/s 50
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 8 of 10 (Start time: 02:51:19 AM, 07 February 2022)

The cells adjacent to the chorda-mesoderm, the paraxial mesoderm cells, are the precursors of the in __

Select the correct option

- Exodermal somites
- Mesodermal somites
- Somites
- Endodermal somites

Solved by Ayesha Meraj

2:51 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 7 of 10 (Start time: 02:50:40 AM, 07 February 2022)

The bird egg is ---------- with a small disc of cytoplasm sitting atop a l

Select the correct option

- Hololecithal
- Isolecithal
- Mesolecithal
- Telolecithal

Solved by Ayesha Meraj

2:50 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 6 of 10 (Start time: 02:49:57 AM, 07 February 2022)

The primitive streak is the major structural characteristic of----------

Select the correct option

- Arthropods
- Mollusks
- Birds
- Avian

Solved by Ayesha Meraj

2:49 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 5 of 10 (Start time: 02:49:17 AM, 07 February 2022)

The cells of both the epiblast and hypoblast intercalate on the future dorsal side of the embryo to form a localized

Select the correct option

- Extension
- Convergence
- Involution
- Embryonic Shield

Solved by Ayesha Meraj

2:49 • 51
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 4 of 10 (Start time: 02:48:47 AM, 07 February 2022)

---------- promotes the formation of MPF, and MPF promotes mitosis.

Select the correct option

- Cyclin synthesis
- Cleavage
- Mitotic division
- Blastomere cell cycle

Solved by Ayesha Meraj

2:48 462 B/s 51
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 3 of 10 (Start time: 02:48:14 AM, 07 February 2022)

Between the area pellucida and the area opaca is a thin layer of cells called the ----------

Select the correct option

- Egg white
- Marginal belt
- Sub germinal cavity
- Albumin

Solved by Ayesha Meraj

2:48 • 529 B/s 51
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

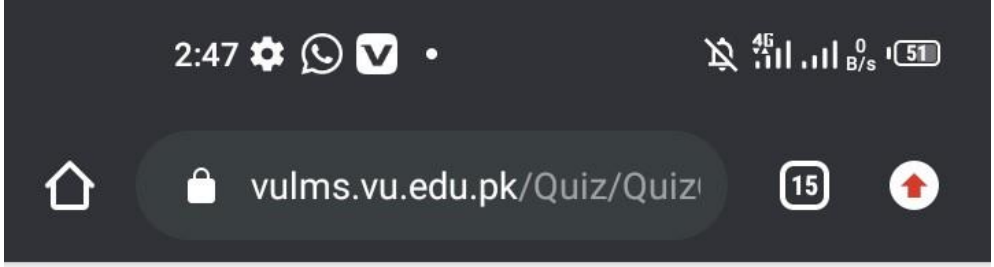
Question # 2 of 10 (Start time: 02:47:17 AM, 07 February 2022)

The ---------- consists of a small group of internal cells surrounded by a larger group of external cells

Select the correct option

- Blastula
- Morula
- Marginal belt
- Marginal zone

Solved by Ayesha Meraj



ZOO501 - Developmental Biology (Quiz # 2)

Question # 1 of 10 (Start time: 02:45:56 AM, 07 February 2022)

The center of this node contains a funnel-shaped depressionthrough which cells can pass into the blastocoels called

Select the correct option

- Primitive streak
- Primitive pit
- Hensen's node
- Primitive knot

Solved by Ayesha Meraj

Click

2:41 342 B/s 52
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 10 of 10 (Start time: 02:40:01 AM, 07 February 2022)

The cells adjacent to the chorda-mesoderm, the paraxial mesoderm cells, are the precursors of the in _

Select the correct option

- (●) Mesodermal somites
- ( ) Exodermal somites
- ( ) Somites
- ( ) Endodermal somites

Solved by Ayesha Meraj

2:38 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 7 of 10 (Start time: 02:38:20 AM, 07 February 2022)

A ---------- is an embryonic structure composed of blastomeres.

Select the correct option

- Blastocyst
- Blastocyte
- Morulla
- Blastula

Solved by Ayesha Meraj

2:37 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 5 of 10 (Start time: 02:36:39 AM, 07 February 2022)

The ---------- is governed by the synthesis and degradation of cyclin.

Select the correct option

- Mesentoblastic cell cycle
- Blastomere cell cycle
- Blastocyte cell cycle
- Blastocyst cell cycle

Solved by Ayesha Meraj

2:36 • 53
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 4 of 10 (Start time: 02:36:04 AM, 07 February 2022)

Tunicate gastrulation is characterized by the invagination of the endoderm, the involution of the -----

Select the correct option

- (●) Mesoderm
- ( ) Endocrine
- ( ) Ectoderm
- ( ) Endoderm

Solved by Ayesha Meraj

2:35

ZOO501 - Developmental Biology (Quiz # 2)

Question # 3 of 10 (Start time: 02:35:19 AM, 07 February 2022)

Pairs of blastomeres appear to lead this invagination into the centre of the embryo. This invagination forms a blasto

Select the correct option

- Endodermal Cells
- Endocrine glands
- Ectodermal Cells
- Mesodermal cells

Clic

Solved by Ayesha Meraj

2:35 • 53
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 2 of 10 (Start time: 02:34:29 AM, 07 February 2022)

While gastrulation in the avian embryo ----------serves as an opening through w

Select the correct option

- Posterior primitive streaks
- Primitive streaks
- Anterior primitive streaks
- Primitive grooves

Solved by Ayesha Meraj

2:34 • 54
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 1 of 10 (Start time: 02:33:52 AM, 07 February 2022)

A mammalian blastula is called a ----------.

Select the correct option

- Blastocyte
- Blastocyts
- Mesentoblast
- Blastomere

Solved by Ayesha Meraj

2:32 484 B/s 54
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 10 of 10 (Start time: 02:32:11 AM, 07 February 2022)

At the anterior end of the primitive streak is a regional thickening of cells called th

Select the correct option

- ○ Primitive streak
- ● Primitive knot
- ○ Primitive groove
- ○ Primitive pit

Solved by Ayesha Meraj

2:32 445 B/s 54
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 9 of 10 (Start time: 02:31:33 AM, 07 February 2022)

The narrow connecting stalk of extra-embryonic ---------- that links the embryo to the

Select the correct option

- (●) Mesoderm
- ( ) Mesentroderm
- ( ) Ectoderm
- ( ) Endoderm

Solved by Ayesha Meraj

2:31 • 4G 0 B/s 54
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 8 of 10 (Start time: 02:30:18 AM, 07 February 2022)

A mammalian blastula is called a ----------.

Select the correct option

- (●) Blastocyts
- ( ) Mesentoblast
- ( ) Blastomere
- ( ) Blastocyte

Solved by Ayesha Meraj

2:30 388 B/s 54
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 7 of 10 (Start time: 02:29:47 AM, 07 February 2022)

In Snail cleavage is ----------

Select the correct option

- (●) Spiral
- ( ) Radial
- ( ) Rotational
- ( ) Bilateral

Solved by Ayesha Meraj

2:29 358 B/s 55
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 5 of 10 (Start time: 02:28:23 AM, 07 February 2022)

The second step of gastrulation in Tunicates involves the involution of the----------

Select the correct option

- Endoderm
- Endocrine
- Mesoderm
- Ectoderm

Solved by Ayesha Meraj

2:28 • 4G 291 B/s 55
vulms.vu.edu.pk/Quiz/Quiz 15

Question # 4 of 10 (Start time: 02:27:21 AM, 07 February 2022)

While gastrulation of the avian embryo the two-layered blastoderm (epiblast and hypoblas the layers forms a ----------

Select the correct option

- (●) Blastocoel
- ( ) Mesoderm
- ( ) Blastoderm
- ( ) Blastocyte

Solved by Ayesha Meraj

2:27 • 4G 151 B/s 55
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 3 of 10 (Start time: 02:26:23 AM, 07 February 2022)

in mammals oocyte is related from ovary and swept by _____into oviduct.

Select the correct option

- fiber
- fimbriae
- both a and b
- none of these

Solved by Ayesha Meraj

2:26 • 342 B/s 55
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 2 of 10 (Start time: 02:25:41 AM, 07 February 2022)

Between the area pellucida and the area opaca is a thin layer of cells called the _____ or marginal

Select the correct option

- (●) Marginal zone
- ( ) Peripheral zone
- ( ) Side zone
- ( ) Interior zone

Solved by Ayesha Meraj

2:25 • 463 B/s 55
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 1 of 10 (Start time: 02:25:07 AM, 07 February 2022)

In mammalian cleavage at 64-cell stage, the inner cell mass and the ---------- cells l

Select the correct option

- ○ Blastocyst
- ○ Cavitation
- ○ Mesentoblast
- ● Trophoblast

Solved by Ayesha Meraj

2:22 409 B/s 56
vulms.vu.edu.pk/Quiz/Quiz 15

MC200404406: MUHAMMAD KASHIF MURTZA

ZOO501 - Developmental Biology (Quiz # 2)

Question # 10 of 10 (Start time: 02:21:16 AM, 07 February 2022)

While gastrulation of the avian embryo the two-layered blastoderm (epiblast and hypoblast) is join the layers forms a ----------

Select the correct option

- Blastoderm
- Mesoderm
- Blastocyte
- Blastocoel

Solved by Ayesha Meraj

2:21 809 B/s 56
vulms.vu.edu.pk/Quiz/Quiz 15

MC200404406: MUHAMMAD KASHIF MURTZA

ZOO501 - Developmental Biology (Quiz # 2)

Question # 9 of 10 (Start time: 02:20:28 AM, 07 February 2022)

The bird egg is ---------- with a small disc of cytoplasm sitting atop a large yolk.

Select the correct option

- Mesolecithal
- Hololecithal
- Telolecithal
- Isolecithal

Solved by Ayesha Meraj

2:20 • 357 B/s 56
vulms.vu.edu.pk/Quiz/Quiz 15

MC200404406: MUHAMMAD KASHIF MURTZA

ZOO501 - Developmental Biology (Quiz # 2)

Question # 8 of 10 (Start time: 02:19:53 AM, 07 February 2022)

In the tunicate cleavage is ----------

Select the correct option

- ○ Radial
- ● Bilateral
- ○ Rotational
- ○ Spiral

Solved by Ayesha Meraj

2:19 • 381 B/s 57
vulms.vu.edu.pk/Quiz/Quiz 15

MC200404406: MUHAMMAD KASHIF MURTZA

ZOO501 - Developmental Biology (Quiz # 2)

Question # 7 of 10 (Start time: 02:19:08 AM, 07 February 2022)

In Birds egg cleavage occurs only in the blastodisc, a small disc of cytoplasm ---------- in c

Select the correct option

- 3-4 mm
- 1-2 mm
- 2-3 mm
- 4-5 mm

Solved by Ayesha Meraj

2:19 • 4G 1 K/s 57
vulms.vu.edu.pk/Quiz/Quiz 15

MC200404406: MUHAMMAD KASHIF MURTZA

ZOO501 - Developmental Biology (Quiz # 2)

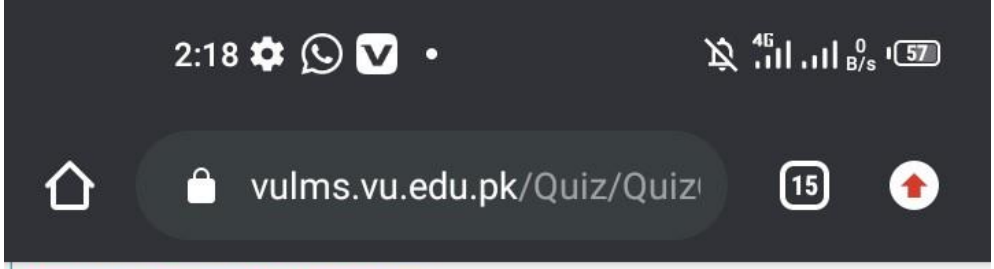
Question # 6 of 10 (Start time: 02:18:24 AM, 07 February 2022)

The yolk syncytial layer is formed at the ----------, when the cells at the vegetal edge of the blasto

Select the correct option

- (●) 9th cell cycle
- ( ) 6th cell cycle
- ( ) 7th cell cycle
- ( ) 8th cell cycle

Solved by Ayesha Meraj



MC200404406: MUHAMMAD KASHIF MURTZA

ZOO501 - Developmental Biology (Quiz # 2) Quiz Start

Question # 5 of 10 (Start time: 02:17:38 AM, 07 February 2022)

Tunicate gastrulation is characterized by the invagination of the endoderm, the involution of the mesoderm, and the epiboly of the

Select the correct option

- Endocrine
- Endoderm
- Mesoderm
- Ectoderm

Click to Save An

Solved by Ayesha Meraj

2:17 • 57
vulms.vu.edu.pk/Quiz/Quiz 15

MC200404406: MUHAMMAD KASHIF MURTZA

ZOO501 - Developmental Biology (Quiz # 2)

Question # 4 of 10 (Start time: 02:17:06 AM, 07 February 2022)

The primitive streak is first visible as a thickening of the ---------- at the posterior region of the embryo, ju

Select the correct option

- (●) Epiblast
- ( ) Mesoblast
- ( ) Mesentoblast
- ( ) Holoblast

Solved by Ayesha Meraj

2:17 • 308 B/s 57
vulms.vu.edu.pk/Quiz/Quiz 15

MC200404406: MUHAMMAD KASHIF MURTZA

ZOO501 - Developmental Biology (Quiz # 2)

Question # 3 of 10 (Start time: 02:16:30 AM, 07 February 2022)

The first cell movement of fish gastrulation is the epiboly of the ---------- cells over the yolk.

Select the correct option

- (●) Blastoderm
- ( ) Ectoderm
- ( ) Mesoderm
- ( ) Endoderm

Solved by Ayesha Meraj

2:16 vulms.vu.edu.pk/Quiz/Quiz 15

MC200404406: MUHAMMAD KASHIF MURTZA

ZOO501 - Developmental Biology (Quiz # 2)

Question # 2 of 10 (Start time: 02:15:40 AM, 07 February 2022)

At the anterior end of the primitive streak is a regional thickening of cells called the

Select the correct option

- Primitive pit
- Primitive knot
- Primitive groove
- Primitive streak

Solved by Ayesha Meraj

2:15 • 533 B/s 57
vulms.vu.edu.pk/Quiz/Quiz 15

MC200404406: MUHAMMAD KASHIF MURTZA

ZOO501 - Developmental Biology (Quiz # 2)

Question # 1 of 10 (Start time: 02:15:08 AM, 07 February 2022)

Cyclin synthesis promotes the formation of ----------

Select the correct option

- FMP
- MFP
- MPF
- PMF

Solved by Ayesha Meraj

2:14 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 10 of 10 (Start time: 02:13:56 AM, 07 February 2022)

In mammalian cleavage at 64-cell stage, the inner cell mass and the ----------

Select the correct option

- Mesentoblast
- Cavitation
- Trophoblast
- Blastocyst

Solved by Ayesha Meraj

2:13 • 95 B/s 58
vulms.vu.edu.pk/Quiz/Quiz 15

The yolk syncytial layer is formed at the ----------, when the cells at the vegetal edge

Select the correct option

- 6th cell cycle
- 8th cell cycle
- 9th cell cycle
- 7th cell cycle

Solved by Ayesha Meraj

2:13 303 B/s 58
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 9 of 10 (Start time: 02:12:38 AM, 07 February 2022)

The yolk syncytial layer is formed at the ----------, when the cells at the vegetal edge of the blastoderm f

Select the correct option

- 6th cell cycle
- 8th cell cycle
- 9th cell cycle
- 7th cell cycle

Solved by Ayesha Meraj

Click

2:12 • 370 B/s 58
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 8 of 10 (Start time: 02:11:56 AM, 07 February 2022)

It will induce the uterine cells of mammals to form the maternal portion of the placenta, the __

Select the correct option

- Nurocidua
- Decidua
- Xcidua
- Lecidua

Solved by Ayesha Meraj

2:11 • 419 B/s 58
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 7 of 10 (Start time: 02:11:07 AM, 07 February 2022)

In the tunicate cleavage is ----------

Select the correct option

- Radial
- Rotational
- Bilateral
- Spiral

Solved by Ayesha Meraj

2:11 • 58
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 6 of 10 (Start time: 02:10:24 AM, 07 February 2022)

Between the area pellucida and the area opaca is a thin layer of cells called the ----

Select the correct option

- Egg white
- Albumin
- Sub germinal cavity
- Marginal belt

Solved by Ayesha Meraj

2:10 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 5 of 10 (Start time: 02:10:02 AM, 07 February 2022)

Mother provides the fetus with nutrients and oxygen, and the fetus sends its waste products ----------

Select the correct option

- ○ Carbon monoxide and urea
- ● Carbon dioxide and urea
- ○ Oxygen and urea
- ○ Urea and oxygen

Solved by Ayesha Meraj

2:09 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 4 of 10 (Start time: 02:09:14 AM, 07 February 2022)

The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the ---

Select the correct option

- Marginal belt
- Area opaca
- Area pellucid
- Marginal zone

Solved by Ayesha Meraj

2:09 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 3 of 10 (Start time: 02:08:30 AM, 07 February 2022)

In the sea urchin, cleavage is ----------

Select the correct option

- (●) Radial
- ( ) Rotational
- ( ) Bilateral
- ( ) Spiral

Solved by Ayesha Meraj

2:08 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 2 of 10 (Start time: 02:07:51 AM, 07 February 2022)

---------- promotes the formation of MPF, and MPF promotes mitosis.

Select the correct option

- Mitotic division
- Cyclin synthesis
- Blastomere cell cycle
- Cleavage

Solved by Ayesha Meraj

2:07 • 357 B/s 59
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 1 of 10 (Start time: 02:06:53 AM, 07 February 2022)

This streak is first visible in birds as a thickening of the _____at the posterior region of the e

Select the correct option

- ○ Mesoblast
- ○ Hypoblast
- ○ Endoblast
- ● Epiblast

Solved by Ayesha Meraj

2:05 • 386 B/s 59
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

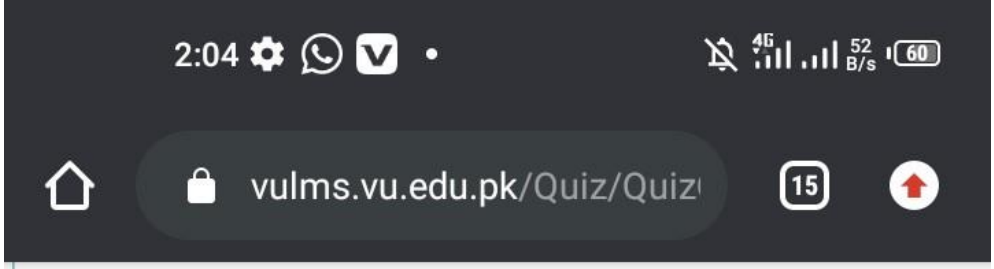
Question # 10 of 10 (Start time: 02:04:41 AM, 07 February 2022)

In mammalian cleavage at 64-cell stage, the inner cell mass and the ---------- cells have t

Select the correct option

- (●) Trophoblast
- ( ) Blastocyst
- ( ) Cavitation
- ( ) Mesentoblast

Solved by Ayesha Meraj



ZOO501 - Developmental Biology (Quiz # 2) Quiz Start Tir

Question # 9 of 10 (Start time: 02:03:45 AM, 07 February 2022)

Tunicate gastrulation is characterized by the invagination of the endoderm, the involution of the mesoderm, and the epiboly of the ---

Select the correct option

- Ectoderm
- Endoderm
- Endocrine
- Mesoderm

Click to Save Answ

Solved by Ayesha Meraj

2:03 • 398 B/s 60
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

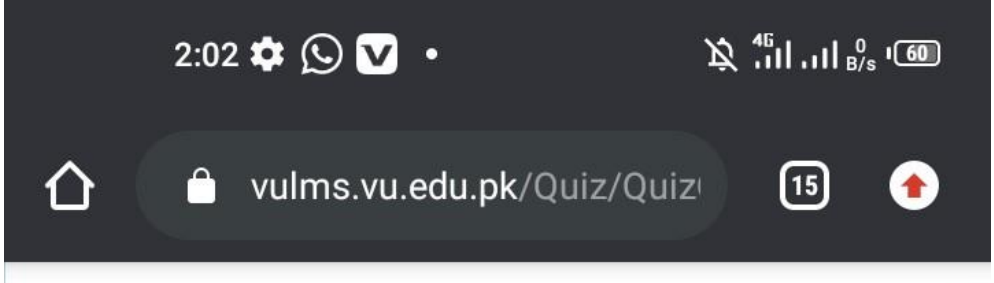
Question # 8 of 10 (Start time: 02:02:59 AM, 07 February 2022)

About ---------- hours after fertilization, the vegetal (endoderm) cells assume a wedge shape, expanding the margins.

Select the correct option

- 4 to 7
- 4 to 6
- 4 to 8
- 4 to 5

Solved by Ayesha Meraj



ZOO501 - Developmental Biology (Quiz # 2) Quiz

Question # 7 of 10 (Start time: 02:01:53 AM, 07 February 2022)

Only the cytoplasm of the blastodisc becomes the embryo, this type of meroblastic cleavage is called discoidal present in .

Select the correct option

- Mammales
- Birds
- Reptile
- Fishes

Click to So

Solved by Ayesha Meraj

2:01 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 6 of 10 (Start time: 02:01:00 AM, 07 February 2022)

The cells adjacent to the chordamesoderm, the paraxial ---------- cells, are the precurs

Select the correct option

- (●) Mesoderm
- ( ) Blastoderm
- ( ) Ectoderm
- ( ) Endoderm

Solved by Ayesha Meraj

2:00 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 5 of 10 (Start time: 02:00:19 AM, 07 February 2022)

In fish eggs, cleavage occurs only in the ---------- a thin region of yolk-free cytoplasm at the c

Select the correct option

- Mesodisk
- Neurotodic
- Holodisc
- Blastodisc

Solved by Ayesha Meraj

2:00 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 4 of 10 (Start time: 01:59:25 AM, 07 February 2022)

Mother provides the fetus with nutrients and oxygen, and the fetus sends its waste products --------

Select the correct option

- Carbon monoxide and urea
- Oxygen and urea
- Urea and oxygen
- Carbon dioxide and urea

Solved by Ayesha Meraj

1:59 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 3 of 10 (Start time: 01:58:39 AM, 07 February 2022)

The dorsal – ventral axis mean

Select the correct option

- Left to right
- Back to belly
- Head to tail
- Mouth to anus

Solved by Ayesha Meraj

1:58 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 2 of 10 (Start time: 01:57:38 AM, 07 February 2022)

The cell divisions do not completely divide the egg, so this type of cleavage is called----------

Select the correct option

- (●) Mesoblast
- ( ) Blastocyte
- ( ) Holoblast
- ( ) Mesentoblast

Solved by Ayesha Meraj

1:57 • 391 B/s 61
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 1 of 10 (Start time: 01:57:05 AM, 07 February 2022)

The mammalian embryo obtains nutrients directly from its ----------

Select the correct option

- (●) Mother
- ( ) Animal pole
- ( ) Stored yolk
- ( ) Vegatal pole

Solved by Ayesha Meraj

1:40 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 10 of 10 (Start time: 01:39:13 AM, 07 February 2022)

In the sea urchin, cleavage is ----------

Select the correct option

- Rotational
- Radial
- Spiral
- Bilateral

Solved by Ayesha Meraj

1:37 • 65
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 8 of 10 (Start time: 01:37:07 AM, 07 February 2022)

During ---------- most cells do not grow. Rather, the volume of the oocyte is cleaved int

Select the correct option

○	Mitosis
○	Meiosis
●	Cleavage
○	Cell cycle

Solved by Ayesha Meraj

1:37 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 7 of 10 (Start time: 01:36:32 AM, 07 February 2022)

Cyclin synthesis promotes the formation of ----------

Select the correct option

- (●) MPF
- ( ) PMF
- ( ) FMP
- ( ) MFP

Solved by Ayesha Meraj

1:36 • 4G 76 B/s 65
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 6 of 10 (Start time: 01:35:10 AM, 07 February 2022)

In the tunicate cleavage is ----------

Select the correct option

- (●) Bilateral
- ( ) Rotational
- ( ) Spiral
- ( ) Radial

Solved by Ayesha Meraj

1:36 • 274 B/s 65
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 6 of 10 (Start time: 01:35:10 AM, 07 February 2022)

In the tunicate cleavage is ----------

Select the correct option

- (●) Bilateral
- ( ) Rotational
- ( ) Spiral
- ( ) Radial

Solved by Ayesha Meraj

1:35 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

Question # 5 of 10 (Start time: 01:34:37 AM, 07 February 2022)

Tunicate gastrulation is characterized by the invagination of the ----------

Select the correct option

- (●) Endoderm
- ( ) Ectoderm
- ( ) Mesoderm
- ( ) Endocrine

Solved by Ayesha Meraj

1:34 379 B/s 65
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2) Quiz Start

Question # 4 of 10 (Start time: 01:33:44 AM, 07 February 2022)

The narrow connecting stalk of extra-embryonic ---------- that links the embryo to the trophoblast eventually forms the ves

Select the correct option

- Mesoderm
- Mesentroderm
- Endoderm
- Ectoderm

Click to Save An

Solved by Ayesha Meraj

1:33 451 B/s 66
vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2) Quiz S

Question # 3 of 10 (Start time: 01:32:29 AM, 07 February 2022)

The cells adjacent to the chorda-mesoderm, the paraxial mesoderm cells, are the precursors of the in _____fish.

Select the correct option

- Exodermal somites
- Mesodermal somites
- Somites
- Endodermal somites

Click to Sav

Solved by Ayesha Meraj

1:32 vulms.vu.edu.pk/Quiz/Quiz 15

ZOO501 - Developmental Biology (Quiz # 2)

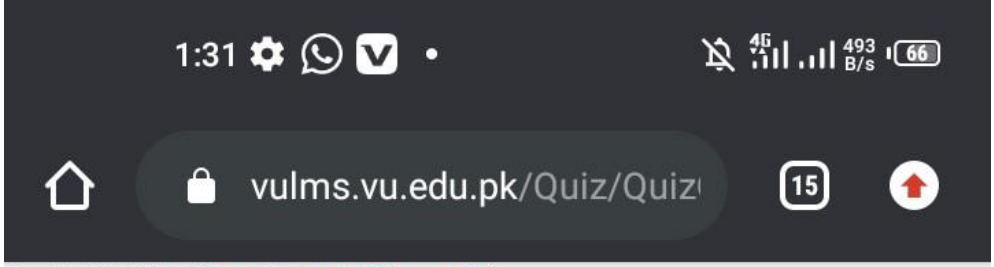
Question # 2 of 10 (Start time: 01:31:47 AM, 07 February 2022)

A ---------- is an embryonic structure composed of blastomeres.

Select the correct option

- Morulla
- Blastocyst
- Blastula
- Blastocyte

Solved by Ayesha Meraj



Question # 1 of 10 (Start time: 01:30:42 AM, 07 February 2022)

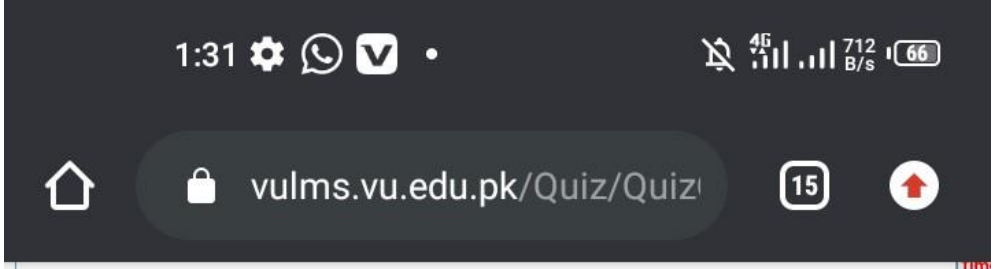
Pairs of blastomeres appear to lead this invagination into the centre of the embryo. This invagination forms a blastopore whose lips will become

Select the correct option

- ○ Ectodermal Cells
- ○ Endocrine glands
- ○ Endodermal Cells
- ● Mesodermal cells

Click to Save Answer & Move

Solved by Ayesha Meraj



ZOO501 - Developmental Biology (Quiz # 2)

Quiz Start Time: 01:30 AM, 07

Question # 1 of 10 (Start time: 01:30:42 AM, 07 February 2022)

Pairs of blastomeres appear to lead this invagination into the centre of the embryo. This invagination forms a blastopore whose lips will become the -----

Select the correct option

- Ectodermal Cells
- Endocrine glands
- Endodermal Cells
- Mesodermal cells

Click to Save Answer & Move to Nex

Solved by Ayesha Meraj

QUIZ 3

FALL 21

ZOO501

VU HELPZONE

GROUO OF ZOOLOGISTA



MARCH 2

DUE DATE

2022

MUHAMMAD KASHIF
03064963933

MC200404406: MUHAMMAD KASHIF MURTZA

ZOO501 - Developmental Biology (Quiz # 3)

Question # 5 of 5 (Start time: 04:02:29 PM, 01 March 2022)

Neurulation in _____ is exclusively secondary.

Solved by Muhammad Kashif

Select the correct option

- (●) Fishes
- ( ) Reptiles
- ( ) Amphibians
- ( ) Mammales

MC200404406: MUHAMMAD KASHIF MURTZA

ZOO501 - Developmental Biology (Quiz # 3) Qu

Question # 4 of 5 (Start time: 04:01:47 PM, 01 March 2022)

Fishes, _____ , and birds undergo discoidal meroblastic cleavage, wherein the early cell divisions do not cut through th

Solved by Muhammad Kashif

Select the correct option

- (●) Reptiles
- ( ) Mammales
- ( ) Fishes
- ( ) Amphibia

MC200404406: MUHAMMAD KASHIF MURTZA

ZOO501 - Developmental Biology (Quiz # 3)

Question # 3 of 5 (Start time: 04:00:17 PM, 01 March 2022)

The region between area opaca and an area pellucida is _____.

Select the correct option

- Central Zone
- Middle Zone
- Marginal Zone
- Intermittent zone

Solved by Muhammad Kashif

MC200404406: MUHAMMAD KASHIF MURTZA

ZOO501 - Developmental Biology (Quiz # 3)

Question # 2 of 5 (Start time: 03:59:51 PM, 01 March 2022)

Fraternal twins are the result of _____ separate fertilization events

Solved by Muhammad Kashif

Select the correct option

- Single
- Two
- Three
- Four

MC200404406: MUHAMMAD KASHIF MURTZA

ZOO501 - Developmental Biology (Quiz # 3) Quiz Start Tir

Question # 1 of 5 (Start time: 03:59:14 PM, 01 March 2022)

The rudiment of the central nervous system, is called _____

Select the correct option

- Neurulation
- Development
- Cell Neurolation
- Differenciation

Solved by Muhammad Kashif

ZOO501 - Developmental Biology (Quiz # 3)

Question # 3 of 5 (Start time: 03:54:57 PM, 01 March 2022)

discoidal meroblastic cleavage occur in _____.

Select the correct option

- mammals
- amphibians
- birds and reptiles
- none of these

Solved by Muhammad Kashif

ZOO501 - Developmental Biology (Quiz # 3) Q

Question # 5 of 5 (Start time: 03:48:26 PM, 01 March 2022)

Gastrulation begins at the

Select the correct option

- (●) Posterior marginal zone
- ( ) Anterior marginal zone
- ( ) Area pellucida
- ( ) Area opaca

Solved by Muhammad Kashif

ZOO501 - Developmental Biology (Quiz # 3)

Question # 4 of 5 (Start time: 03:47:41 PM, 01 March 2022)

Neurulation in _____ is exclusively secondary.

Select the correct option

- Mammales
- Fishes
- Amphibians
- Reptiles

Solved by Muhammad Kashif

ZOO501 - Developmental Biology (Quiz # 3)

Question # 3 of 5 (Start time: 03:47:24 PM, 01 March 2022)

The prechordal _____ induces the formation of the forebrain; the chordamesoderm induces the formation of the

Select the correct option

- (●) Plate
- ( ) Dyer
- ( ) Layer
- ( ) Membrane

Solved by Muhammad Kashif

ZOO501 - Developmental Biology (Quiz # 3)

Question # 2 of 5 (Start time: 03:47:07 PM, 01 March 2022)

Fraternal twins are the result of _____ separate fertilization events

Select the correct option

- (●) Two
- ( ) Three
- ( ) Single
- ( ) Four

Solved by Muhammad Kashif

ZOO501 - Developmental Biology (Quiz # 3)

Question # 1 of 5 (Start time: 03:46:52 PM, 01 March 2022)

The region between area opaca and an area pellucida is _____.

Select the correct option

- Central Zone
- Marginal Zone
- Intermittent zone
- Middle Zone

Solved by Muhammad Kashif

ZOO501 - Developmental Biology (Quiz # 3)

Quiz Start Time: 03:39 PM, 01 Mar

Question # 5 of 5 (Start time: 03:43:01 PM, 01 March 2022)

Total

Neurulation in _____ is exclusively secondary.

Select the correct option

- Mammales
- Reptiles
- Fishes
- Amphibians

Solved by Muhammad Kashif

Sc

ZOO501 - Developmental Biology (Quiz # 3)

Question # 5 of 5 (Start time: 03:43:01 PM, 01 March 2022)

Neurulation in _____ is exclusively secondary.

Select the correct option

- Amphibians
- Fishes
- Reptiles
- Mammales

Solved by Muhammad Kashif

ZOO501 - Developmental Biology (Quiz # 3)

Question # 4 of 5 (Start time: 03:41:44 PM, 01 March 2022)

The region between area opaca and an area pellucida is _____.

Select the correct option

- Intermittent zone
- Central Zone
- Marginal Zone
- Middle Zone

Solved by Muhammad Kashif

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ZOO501 - Developmental Biology (Quiz # 3) **Quiz Start Time: 03:39 PM, 01 March 20**

Question # 3 of 5 (Start time: 03:40:41 PM, 01 March 2022) **Total Mark**

Fraternal twins are the result of _____ separate fertilization events

Select the correct option

- Three
- Four
- Two
- Single

Solved by Muhammad Kashif

ZOO501 - Developmental Biology (Quiz # 3)

Question # 3 of 5 (Start time: 03:40:41 PM, 01 March 2022)

Fraternal twins are the result of _____ separate fertilization events

Select the correct option

○	Three
○	Single
●	Two
○	Four

Solved by Muhammad Kashif

ZOO501 - Developmental Biology (Quiz # 3)

Question # 2 of 5 (Start time: 03:40:12 PM, 01 March 2022)

The rudiment of the central nervous system, is called _____

Select the correct option

- Development
- Neurulation
- Differenciation
- Cell Neurolation

Solved by Muhammad Kashif

ZOO501 - Developmental Biology (Quiz # 3)

Question # 1 of 5 (Start time: 03:39:33 PM, 01 March 2022)

The prechordal _____ induces the formation of the forebrain; the chordamesoderm induces the formation of

Solved by Muhammad Kashif

Select the correct option

- (●) Plate
- ( ) Dyer
- ( ) Layer
- ( ) Membrane